

AWS NETWORKING AND CONTENT DELIVERY

Module 3
BCSE355L – AWS Solution Architect



Agenda

AWS Global Infrastructure

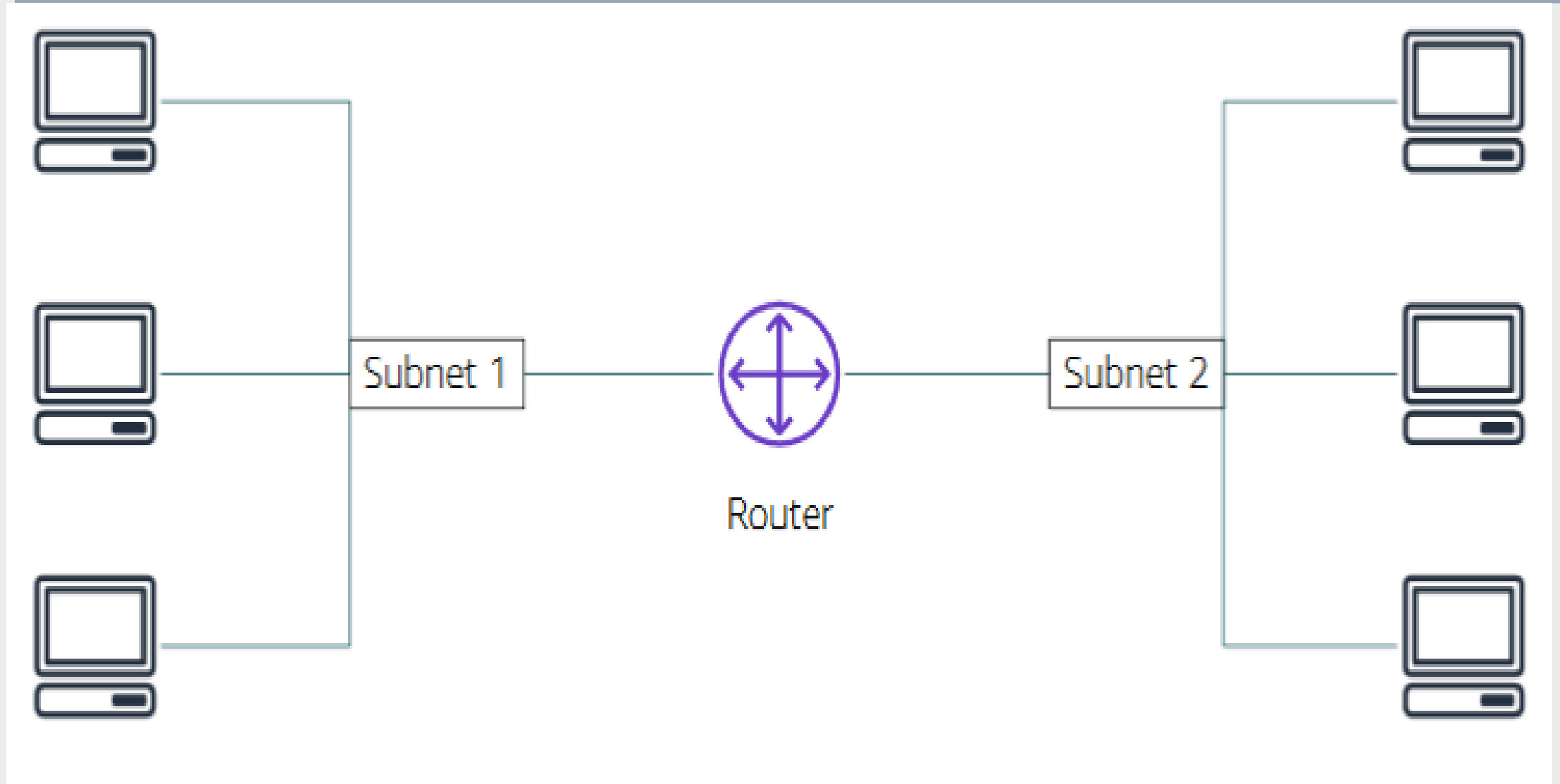
AWS Networking foundations

Edge networking

Networking in AWS

- ❖ Networking is the first step for any organization to set up its landing zone and the entire IT workload built on top of it.
- ❖ Networking is the backbone of the IT application and infrastructure workload.
- ❖ AWS provides various networking services for building your IT landscape in the cloud.
- ❖ With a traditional on-premise IT workload, it becomes challenging to scale globally and provide the same user experience across the globe.
- ❖ AWS **helps solve these problems through edge networking**, and you will learn more about deploying your application for global users without compromising their experience.

Networking in AWS



IP Address

- ❖ Each client machine in a network **has a unique Internet Protocol (IP) address** that identifies it.
- ❖ An **IP address is a numerical label in decimal format.** Machines convert that decimal number to a binary format.
- ❖ In this example, the IP address is 192.0.2.0.
- ❖ Each of the four dot (.)-separated numbers of the IP address represents **8 bits in octal number format.**
- ❖ That means each of the four numbers can be anything from **0 to 255.**
- ❖ The combined total of the four numbers for an IP address is 32 bits in binary format.

Networking in AWS

192	.	0	.	2	.	0
↓		↓		↓		↓
11000000		00000000		00000010		00000000

IPv4 (32-bit) address: 192.0.2.0

IPv6 (128-bit) address:

2600:1f18:22ba:8c00:ba86:a05e:a5ba:00FF

IPv4 and IPv6 addresses

- ❖ A 32-bit IP address is called **an IPv4 address**.
- ❖ IPv6 addresses, which are 128 bits, are also available. IPv6 addresses can accommodate more user devices.
- ❖ An IPv6 address is composed of eight groups of four letters and numbers that are separated by colons (:).
- ❖ In this example, the **IPv6 address is 2600:1f18:22ba:8c00:ba86:a05e:a5ba:00FF**.
- ❖ Each of the **eight colon-separated groups** of the IPv6 address represents 16 bits in hexadecimal number format.
- ❖ That means each of the eight groups can be anything from **0 to FFFF**.
- ❖ The combined total of the eight groups for an **IPv6 address is 128 bits in binary format**.



About AWS Global Infrastructure

About AWS Global Infrastructure

- ❖ The infrastructure offered by AWS is **highly secure and reliable**.
- ❖ It offers **over 200 services**. Most are available in all AWS Regions worldwide, **spread across 245 countries**.
- ❖ Regardless of the type of technology application you are planning to build and deploy, AWS is sure to provide a service that will facilitate its deployment.
- ❖ AWS has millions of customers and thousands of consulting and technology partners worldwide.
- ❖ Businesses large and small across all industries rely on AWS to handle their workloads.

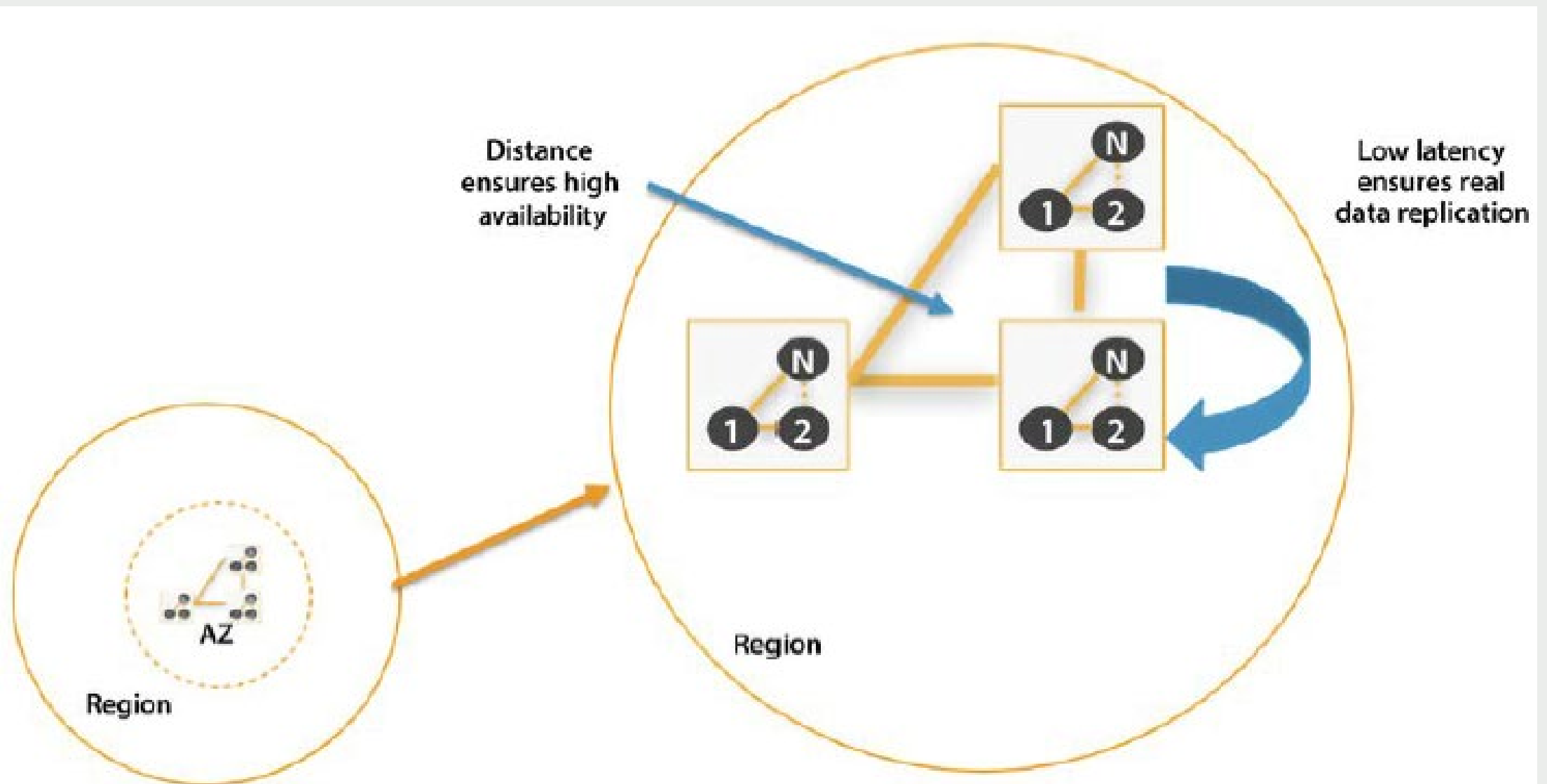
About AWS Global Infrastructure

- ❖ AWS provides the following as its global infrastructure:
 - ❖ 26 launched Regions and 8 announced Regions
 - ❖ 84 Availability Zones
 - ❖ Over 110 Direct Connect locations
 - ❖ Over 310 Points of Presence
 - ❖ 17 Local Zones and 32 announced LZs
 - ❖ 24 Wavelength Zones

AWS Global Infrastructure

- ❖ Amazon provide such a reliable service across the globe?
- ❖ How can they offer reliability and durability guarantees for some of their services?
- ❖ The answer reveals why they are the cloud leaders and why it's difficult to replicate what they offer.
- ❖ AWS has billions of dollars worth of infrastructure deployed across the world. These locations are organized into different **Regions and Zones**.
- ❖ AWS calls them the following:
 - ❖ **AWS Regions**
 - ❖ **Availability Zones (AZs)**
 - ❖ **Local Zones (LZs)**

AWS Global Infrastructure



AWS Global Infrastructure

- ❖ AZ is comprised of multiple distinct data centers, each *equipped with redundant power, networking, and connectivity, and located in separate facilities.*
- ❖ **AWS Regions** exist in separate geographic areas.
- ❖ Each **AWS Region** comprises several independent and isolated data centers (AZs) that provide a full array of AWS services.
- ❖ AWS is continuously enhancing its data centers to provide the latest technology.
- ❖ **AWS's data centers** have a high degree of redundancy.
- ❖ AWS uses highly reliable hardware, but the hardware is not foolproof.

AWS Global Infrastructure

- ❖ Occasionally, a failure can happen that interferes with the availability of resources in each data center.
- ❖ Suppose all instances were hosted in only one data center. If a failure occurred with the whole data center, then none of your resources would be available.
- ❖ AWS mitigates this issue by having multiple data centers in each Region.

AWS Regions

- ❖ AWS Regions **are groups of data centers in one geographic location** - designed to be independent and isolated from each other.
- ❖ A **single Region** consists of **a collection of data centers spread within that Region's geographic boundary** - independence promotes availability and enhances fault tolerance and stability.
- ❖ AWS services available in that Region.
- ❖ There is a possibility that a particular service is not available in your Region.
- ❖ Eventually, all services become **generally available (GA)** after their launch; however, the timing of availability may differ between different Regions.

AWS Regions

- ❖ Eg., **Direct Connect Gateway (DXGW)**, **Identity and Access Management (IAM)**, Cloud-Front, and Route53.
- ❖ Other services are not global but allow you to create inter-Region fault tolerance and availability.
- ❖ For example, Amazon **Relational Database Service (RDS)** allows you to create read replicas in multiple Regions.
- ❖ **Advantages**
 - ❖ Resources will be **closer to users, increasing access speed and reducing latency**.
 - ❖ **Serve your clients without disruption** even if a whole Region becomes unavailable by planning your disaster recovery workload to be in another Region.

AWS AZs

- ❖ AZs are components of AWS Regions.
- ❖ The **clusters of data centers within a Region** are called **AZs**.
- ❖ A single AZ consists of multiple data centers,
 - ❖ are connected to each other using ***AWS-owned dedicated fiber optic cables***
 - ❖ located ***within a 60-mile radius***,
 - ❖ far enough to ***avoid localized failures***,
 - ❖ achieves a ***faster data transfer between the data centers***.
- ❖ AZs have multiple power sources, redundant connectivity, and redundant resources.

AWS AZs

- ❖ The AZs within an AWS Region are interconnected.
- ❖ These connections have the following properties:
 - ❖ Fully redundant
 - ❖ High-bandwidth
 - ❖ Low-latency
 - ❖ Scalable
 - ❖ Encrypted
 - ❖ Dedicated

AWS LZs

- ❖ AZs focus on covering larger areas throughout regions (like highly populated cities) through newer component of AWS infrastructure i.e. **LZs (Local Zones)**.
- ❖ LZs place **select services close to end users**, allowing them to create AWS applications that **deliver single-digit, millisecond responses**.
- ❖ An LZ is the compute and storage infrastructure located close to high-population areas and industrial centers, and offers **high-bandwidth, low-latency connectivity** to the broader AWS infrastructure.
- ❖ Due to their proximity to the customer, LZs facilitate the delivery of applications that necessitate latency in single-digit milliseconds to end-users. **AWS had has 32 LZs in US.**

AWS LZs

- ❖ AWS LZs can run various AWS services, such as
 - ❖ Amazon Elastic Compute Cloud,
 - ❖ Amazon Virtual Private Cloud,
 - ❖ Amazon Elastic Block Store,
 - ❖ Amazon Elastic Load Balancing,
 - ❖ Amazon FSx,
 - ❖ Amazon EMR,
 - ❖ Amazon ElastiCache,
 - ❖ Amazon RDS in geographic proximity to your end users.

Benefits of the AWS Global Infrastructure

- ❖ **Security** - AWS's shared security responsibility model, you offload infrastructure security to AWS and focus on the application security that matters for your business.
- ❖ **Availability** - AWS Regions are fully isolated, and within each Region, the AZs are further isolated partitions of AWS infrastructure. You can use AWS infrastructure with an on-demand model to deploy your applications across multiple AZs in the same Region or any Region globally.
- ❖ **Performance** – Performance is another critical factor in retaining and increasing the user base. AWS provides low-latency network infrastructure by using redundant 100 GbE fiber, which leads to terabits of capacity between regions.

Benefits of the AWS Global Infrastructure

- ❖ **Scalability** – When user demands increase, you must have the **required capacity to scale your application**. With AWS, you can quickly **spin up resources, deploying thousands of servers in minutes to handle any user demand**.
- ❖ **Flexibility** – With AWS, **you can choose how and where to run your workloads**; for example, you can run applications globally by deploying into any of the AWS Regions and AZs worldwide.



AWS Network Foundation

AWS Network Foundations

- ❖ Networking concepts are the same when it comes to the cloud.
- ❖ Know about what networking is, but instead how to set up your private network in the AWS cloud and establish connectivity between the different servers in the cloud and from on-premises to an AWS cloud.
- ❖ First, let's start with the foundation; the first step to building your networking backbone in AWS is using Amazon VPC.

Amazon Virtual Private Cloud (VPC)

- ❖ VPC is one of the core services AWS provides.
- ❖ VPC is your version of the AWS cloud, and as the name suggests, **it is “private,”** which means that by default, your VPC **is a logically isolated and private network inside AWS.**
- ❖ VPC as being the same as your own logical data center in a virtual setting inside the AWS cloud, where you have complete control over the resources inside your VPC.
- ❖ AWS resources like AWS servers, and Amazon EC2 and Amazon RDS instances are placed inside the VPC, including all the required networking components to control the data traffic as per your needs.

Amazon Virtual Private Cloud (VPC)

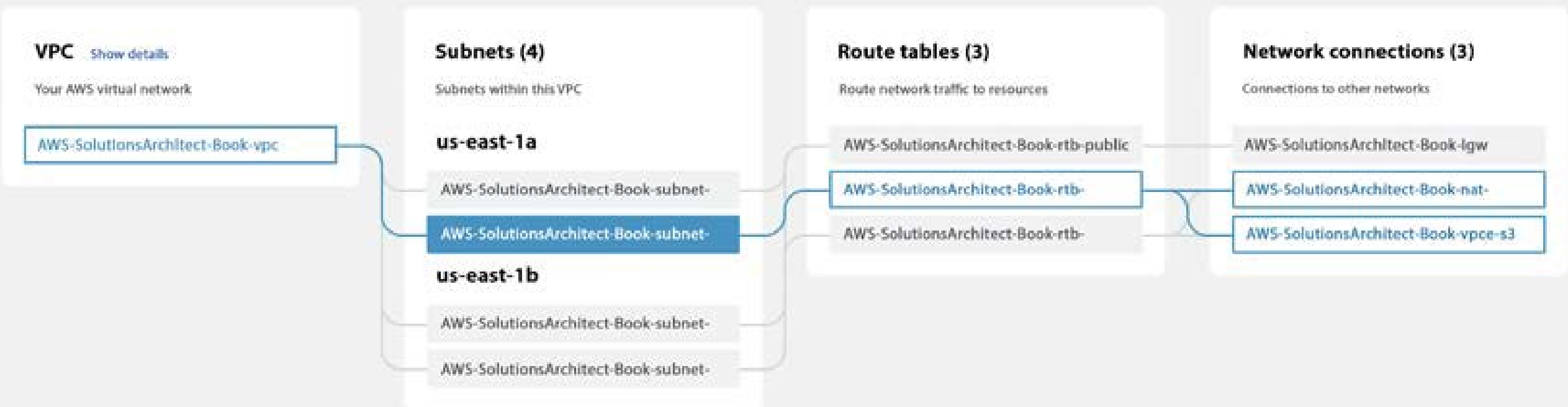


Amazon
VPC

- Enables you to provision a **logically isolated** section of the AWS Cloud where you can launch AWS resources in a virtual network that you define
- Gives you **control over your virtual networking resources**, including:
 - Selection of IP address range
 - Creation of subnets
 - Configuration of route tables and network gateways
- Enables you to **customize the network configuration** for your VPC
- Enables you to use **multiple layers of security**

Amazon Virtual Private Cloud (VPC)

- ❖ Creating a VPC could be a very complex task, but AWS has made it easy by providing Launch VPC Wizard.
- ❖ The following screenshot shows the VPC network configuration across two AZs, us-east-1a and us-east-1b:



Amazon Virtual Private Cloud (VPC)

- ❖ VPCs spread across two AZs, where each AZ has two subnets – one public and one private.
 - ❖ The highlighted flow shows the data flow of a server deployed into a private subnet of the us-east-1 AZ.
 - ❖ Before going into further details, let's look at key VPC concepts to understand them better:
- ❖ Classless Inter-Domain Routing (CIDR) blocks
 - ❖ Subnets
 - ❖ Route tables
 - ❖ Internet Gateway
 - ❖ Network Address Translation (NAT) Gateway
 - ❖ Security Groups (SGs)
 - ❖ Network Access Control List (NACL)
 - ❖ Egress-only IGWs
 - ❖ DHCP
 - ❖ VPC Flow Logs

Amazon Virtual Private Cloud (VPC)

- ❖ **Classless Inter-Domain Routing (CIDR) blocks:** CIDR is the IP address range allocated to your VPC.
- ❖ Creating a VPC, specify its set of IP addresses with CIDR notation.
- ❖ CIDR notation is a simplified way of showing a specific range of IP addresses.
- ❖ For example, **10.0.0.0/16 covers all IPs from *10.0.0.0 to 10.0.255.255, providing 65,535 IP addresses to use.***
- ❖ All resources in your VPC must fall within the CIDR range.

Classless Inter-Domain Routing (CIDR)

Network identifier (routing prefix)

192 . 0 . 2

Host identifier

. 0 / 24



11000000

Fixed



00000000

Fixed



00000010

Fixed



00000000
to 11111111

Flexible

Tells you how
many bits are
fixed

Open System Interconnection (OSI) Model

Layer	Number	Function	Protocol/Addresses
Application	7	Means for an application to access a computer network	HTTP(S), FTP, DHCP, LDAP
Presentation	6	- Ensures that the application layer can read the data - Encryption	ASCII, ICA
Session	5	Enables orderly exchange of data	NetBIOS, RPC
Transport	4	Provides protocols to support host-to-host communication	TCP, UDP
Network	3	Routing and packet forwarding (routers)	IP
Data link	2	Transfer data in the same LAN network (hubs and switches)	MAC
Physical	1	Transmission and reception of raw bitstreams over a physical medium	Signals (1s and 0s)

Amazon Virtual Private Cloud (VPC)

❖ Subnets:

- ❖ Subnet is the VPC CIDR block subset. Partitions of the network are divided by the CIDR range within the range of IP addresses in your VPC.
- ❖ A VPC can have multiple subnets for different kinds of services or functions, like a frontend subnet (for internet access to a web page), a backend subnet (for business logic processing), and a database subnet (for database services).
- ❖ Subnets create trusted boundaries between private and public resources, organize your subnets based on internet accessibility, allows to define clear isolation between public and private resources.
- ❖ Majority of resources can be hosted in private subnets.

Amazon Virtual Private Cloud (VPC)

- ❖ **Route tables:** A routing table contains a set of rules called **routes**. Routes **determine where the traffic will flow**.
- ❖ By default, **every subnet has a routing table**.
- ❖ Manually create a new route table and assign subnets to it. For better security, use the custom route table for each subnet.
- ❖ **An Internet Gateway (IGW):** The IGW sits at the edge of the VPC and **provides connectivity between your VPC resources and the public network** (the internet).
- ❖ By default, internet accessibility is denied for internet traffic in your environment.
- ❖ An IGW **needs to be attached to your public subnet through the subnet's route table**, defining the rules to the IGW.

Amazon Virtual Private Cloud (VPC)

- ❖ **Network Address Translation (NAT) gateways:**
- ❖ A NAT gateway provides outbound internet access to the private subnet and prevents connections from being initiated from outside to your VPC resources.
- ❖ A private subnet blocks all incoming and outgoing internet traffic, but servers may need outgoing internet traffic for software and security patch installation.
- ❖ A NAT gateway enables instances in a private subnet to initiate outbound traffic to the internet and protects resources from incoming internet traffic.
- ❖ All restricted servers (such as database and application resources) should deploy inside your private subnet.

Amazon Virtual Private Cloud (VPC)

❖ Security Groups (SGs):

- ❖ SGs are the virtual firewalls for your instances to control inbound and outbound packets.
- ❖ Only allow statements can used in the SG, and everything else is denied implicitly.
- ❖ SGs control inbound and outbound traffic as designated resources for one or more instances from the CIDR block range or another SG.

Amazon Virtual Private Cloud (VPC)

- ❖ **Network Access Control List (NACL):**

- ❖ A NACL is **another firewall that sits at the subnet boundary** and **allows or denies incoming and outgoing packets**.
- ❖ The main difference between a NACL and an SG is that the **NACL is stateless** – therefore, you **need to have rules for incoming and outgoing traffic**.
- ❖ With an SG, **you need to allow traffic in one direction, and return traffic is, by default, allowed**.
- ❖ **SG** in most places as it is a **firewall at the EC2 instance level**, while a **NACL is a firewall at the subnet level**.
- ❖ **NACL can put control at the VPC level and also deny specific IPs**, as an SG cannot have a deny rule for network traffic coming for a particular IP or IP range.

Amazon Virtual Private Cloud (VPC)

- ❖ **Egress-only IGWs:**

- ❖ It **provide outbound communication from Internet Protocol version 6 (IPv6) instances** in your VPC to the internet and prevent the inbound connection from the internet to your instances on IPv6.

- ❖ IPv6, the sixth iteration of the Internet Protocol, succeeds IPv4 and employs a 128-bit IP address. Like IPv4, it facilitates the provision of unique IP addresses required for internet-connected devices to communicate.

- ❖ **DHCP option sets:**

- ❖ This is **a group of network information**, such as DNS name server and domain name used by EC2 instances when they launch.

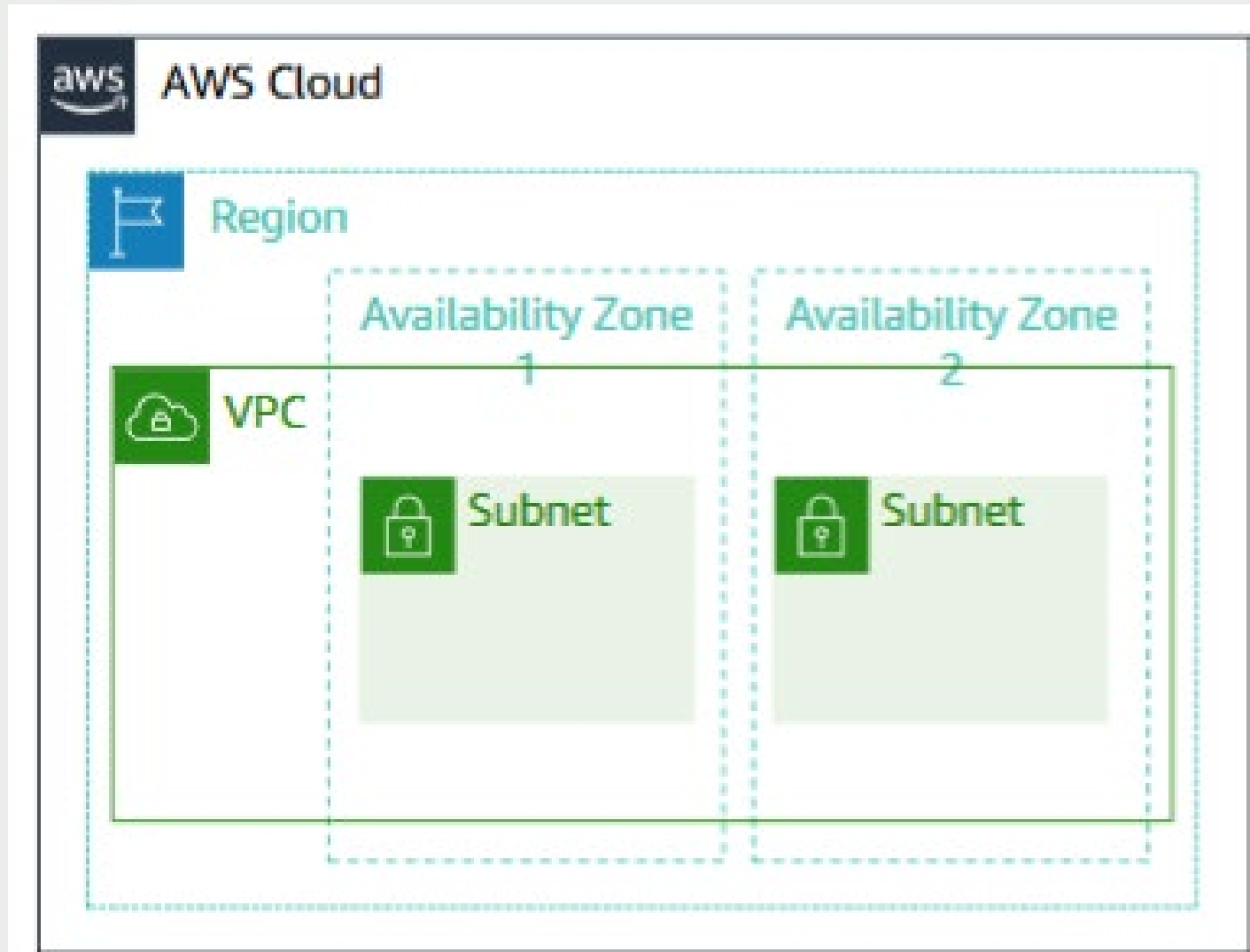
Amazon Virtual Private Cloud (VPC)

❖ VPC Flow Logs:

- ❖ These **enable you to monitor traffic flow to your system VPC**, such as accepted and rejected traffic information for the designated resource to understand traffic patterns.
- ❖ Flow Logs can also be **used as a security tool for monitoring traffic reaching your instance.**
- ❖ **Can create alarms to notify** you if certain types of traffic are detected.

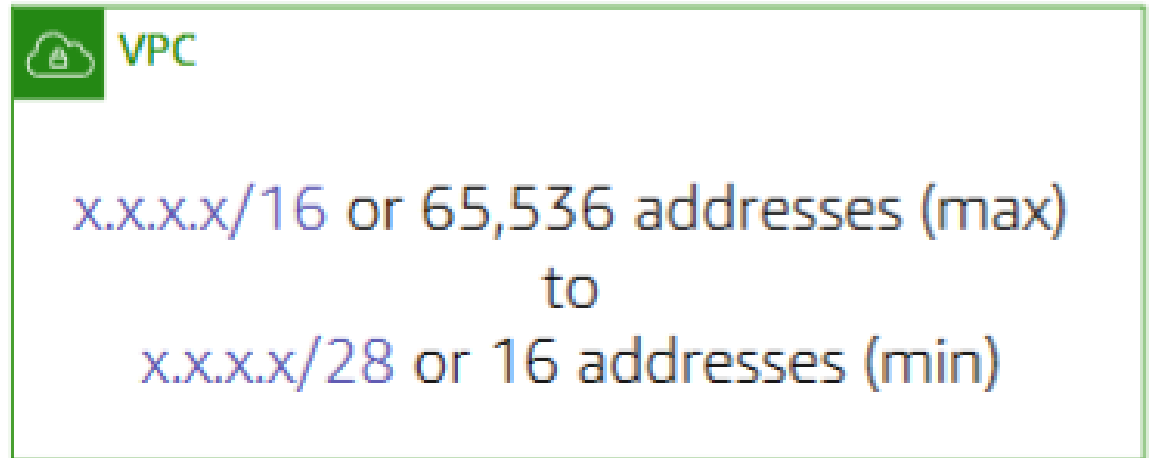
Amazon VPC

- ❖ **VPCs:**
- ❖ Logically isolated from other VPCs.
- ❖ Dedicated to your AWS account
- ❖ Belong to a single AWS Region and can span multiple AZ
- ❖ **Subnets:**
- ❖ Range of IP addresses that divide a VPC.
- ❖ Belong to a single AZ
- ❖ Classified as public or private.



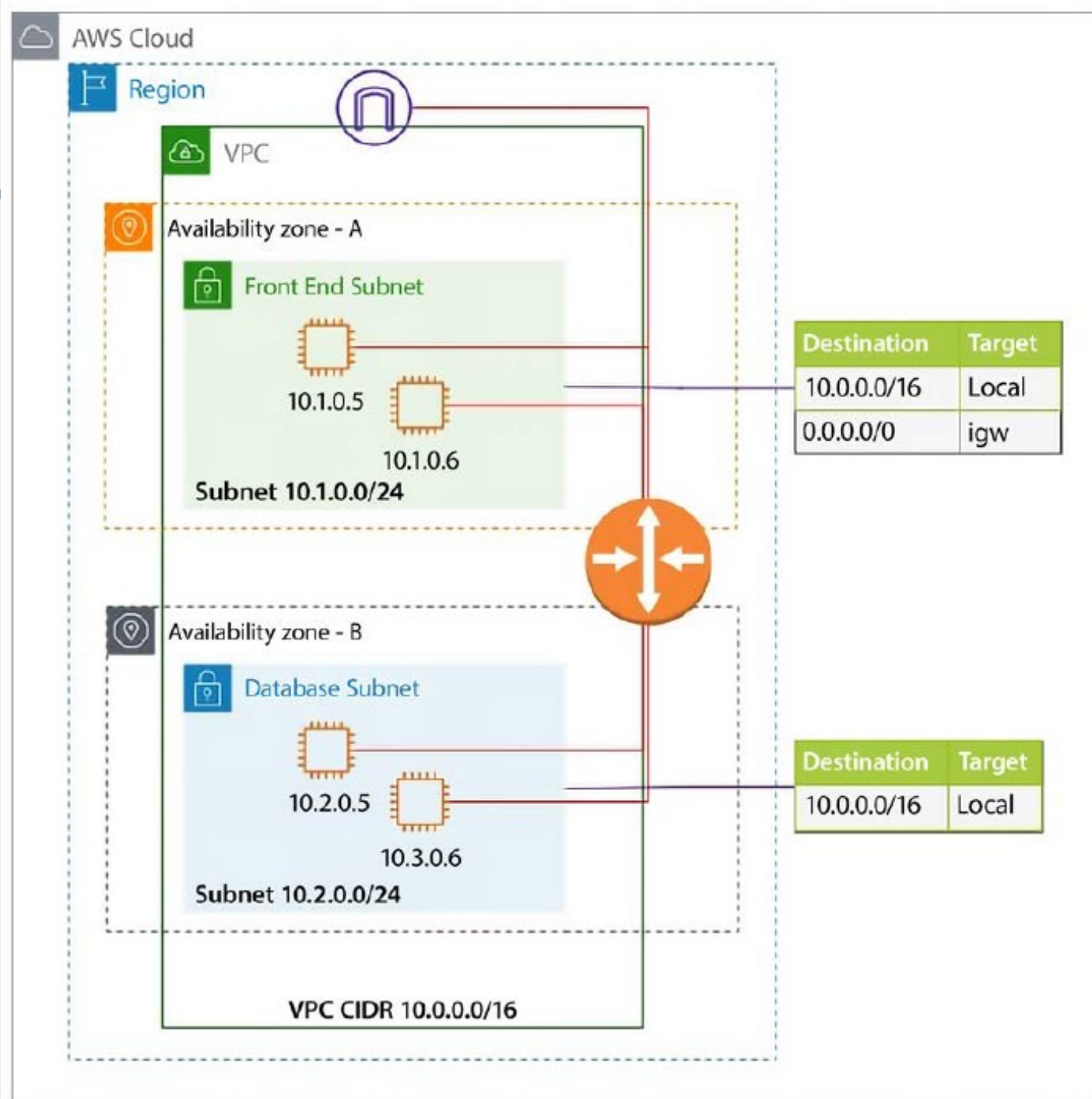
Amazon VPC

- ❖ **IP addressing**
- ❖ When you create a VPC, assign to an IPv4 **CIDR block**.
- ❖ **Cannot change the address range** after you create the VPC.
- ❖ **Largest IPv4** CIDR block size is /16.
- ❖ **Smallest IPv4** CIDR block size is /28.
- ❖ IPv6 is also supported.
- ❖ CIDR blocks of subnets **cannot overlap**.



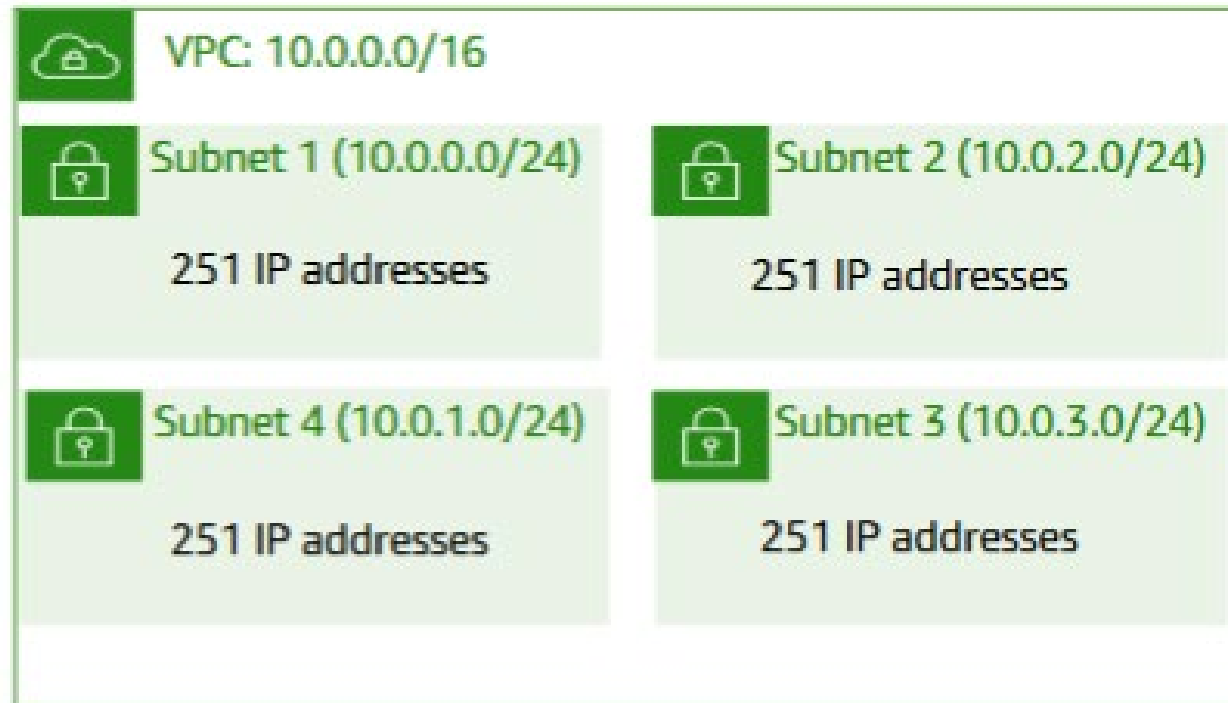
Amazon VPC

- ❖ VPC subnets can be **either private or public.**
- ❖ Name suggests, a private subnet doesn't have access to and from the internet, and a public subnet does.
- ❖ By default, **any subnet you create is private; what makes it public is the default route – as in, 0.0.0.0/0, via the IGW.**



Reserved IP Addresses

- ❖ Example: A VPC with an IPv4 CIDR block of 10.0.0.0/16 has 65,536 total IP addresses. The VPC has 4 equal-sized subnets. Only 251 IP addresses are available for use by each subnet.



IP Addresses for CIDR block 10.0.0.0/24	Reserved for
10.0.0.0	Network address
10.0.0.1	Internal communication
10.0.0.2	Domain Name System (DNS) resolution
10.0.0.3	Future use
10.0.0.255	Network broadcast address

Public IP Addresses

- ❖ **Public IPv4 Address**

- ❖ **Manually assigned through an Elastic IP address**

- ❖ Automatically assigned through the auto-assign public IP address setting at the subnet level.

- ❖ **Elastic IP Address**

- ❖ Associated with an AWS account
- ❖ Can be allocated and remapped anytime
- ❖ Additional costs might apply.

Elastic Network Interface

- ❖ An elastic network interface is a **virtual network interface** that can:
 - ❖ **Attach to an** instance
 - ❖ **Detach from** the instance, and attach to another instance to redirect network traffic.
- ❖ It **attributes follow** when it reattached to a new instance
- ❖ Each instance in your VPC has a **default network interface** that is assigned a private IPv4 address from IPv4 address range of your VPC.



Route table and routes

- ❖ A **route table** contains a **set of rules** that **you can configure** to direct network traffic from your subnet.
- ❖ Each **route** specifies a **destination** and a **target**.
- ❖ By default, every route table contains a **local route** for communication within the VPC.
- ❖ **Each subnet must be associated with a route table.**

Main (Default) Route Table

Destination	Target
10.0.0.0/16	local

VPC CIDR block

Amazon Virtual Private Cloud (VPC)

- ❖ VPC's route tables comprise directives for packet routing, and a default route table exists.
- ❖ Unique route tables can be assigned to individual subnets.
By default, all VPC subnets possess interconnectivity.
- ❖ VPC enhancements for more precise subnet routing, which enables configuration of subnet route tables that direct traffic between two subnets in a VPC through virtual appliances like intrusion detection systems, network firewalls, and protection systems.
- ❖ VPC creation and building tighter security, organizations tend to create multiple VPCs, which makes things more complicated when these VPCs need to communicate with each other. To simplify, we use **Transit Gateway (TGW)**.

AWS Transit Gateway (TGW)

- ❖ Customers are spinning more and more VPCs in AWS, there is an ever-increasing need to connect various VPCs.
- ❖ Before TGW, you could connect VPCs using VPC peering, but VPC peering is a one-to-one connection, which means that resources within peered VPCs only can communicate with each other.
- ❖ If multiple VPCs need to communicate with each other, which is often the case, it results in a complex mesh of VPC peering.
- ❖ For example, as shown in the diagram below, if you have 5 VPCs, you need 10 peering connections.

AWS Transit Gateway (TGW)

- ❖ Managing so many VPC peering connections will become challenging, and there is also a limit on the number of peering connections per account.
- ❖ To overcome this challenge, AWS released TGW.
- ❖ TGW needs one connection called an attachment to a VPC, and you can establish full- or part-mesh connectivity easily without maintaining so many peering connections.

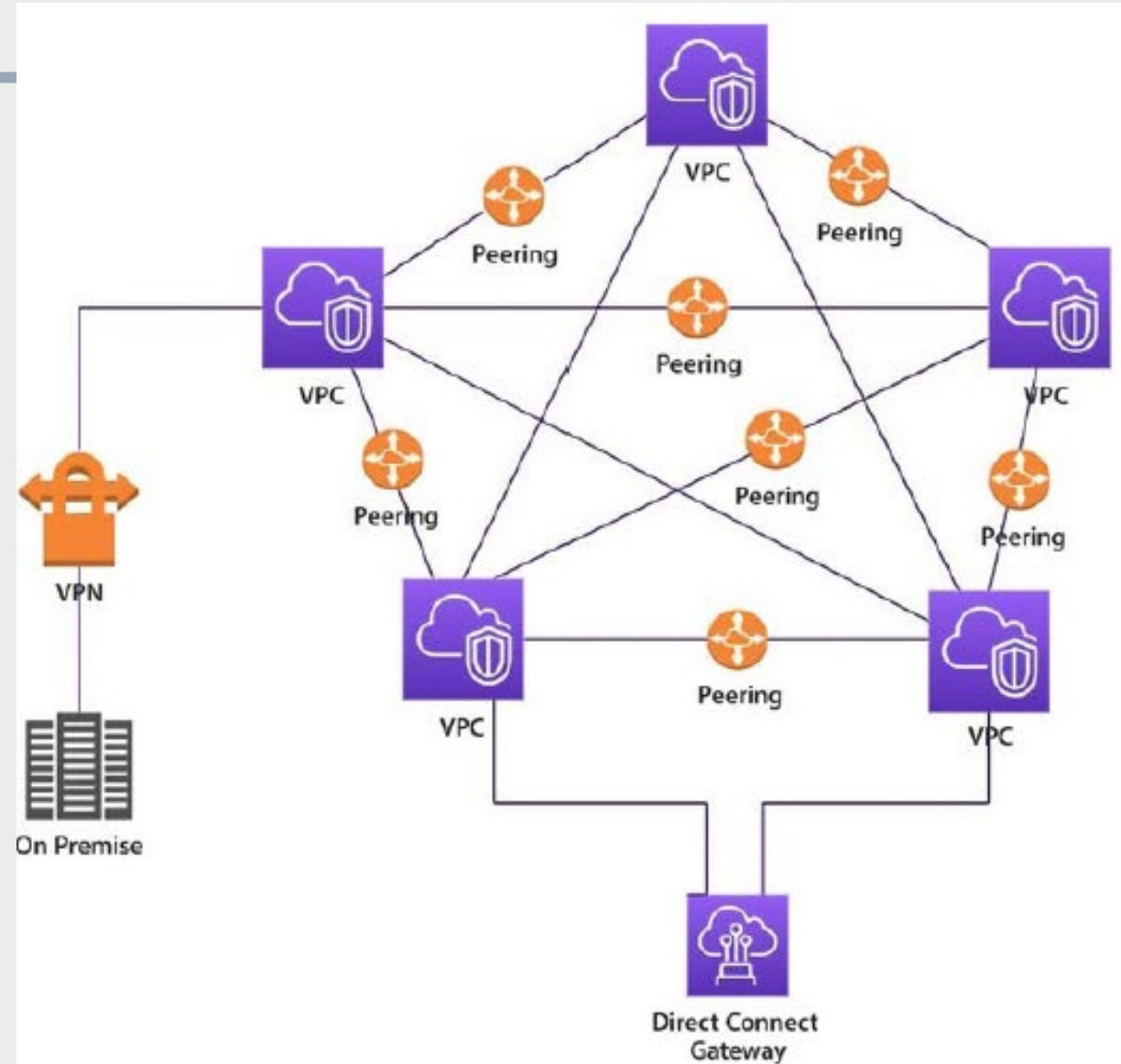


Figure 4.4: VPC connectivity using VPC peering without TGW

AWS Transit Gateway (TGW)

- ❖ Diagram shows simplified communication between five VPCs using TGW.
- ❖ AWS TGW is a **central aggregation service spanned within a Region,** which can be used to connect your VPCs and on-premises networks.
- ❖ TGW is a **managed service that takes care of your availability and scalability and eliminates complex VPN or peering connection scenarios** when connecting with multiple VPCs and on-premises infrastructure.

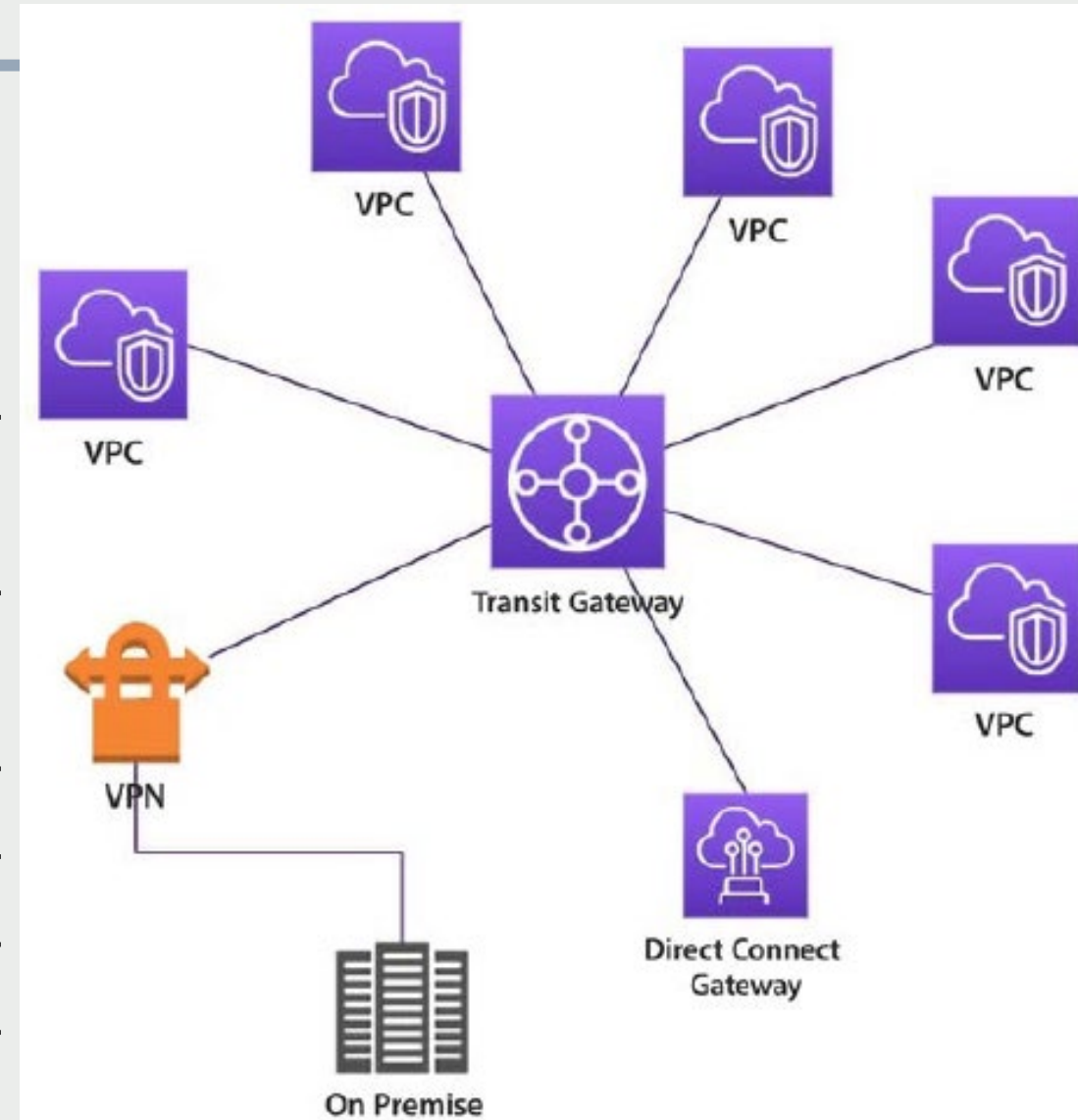


Figure 4.5: VPC connectivity with TGW

AWS Transit Gateway (TGW)

- ❖ You **can connect TGWs in different Regions** by using TGW peering.
- ❖ ***TGW is a Regional entity***, meaning you can only attach VPCs to the TGW within the same Region, and per VPC, the bandwidth reserved is 50 Gbps.
- ❖ However, one TGW can have up to 5,000 VPC attachments.

AWS PrivateLink

- ❖ **AWS PrivateLink** establishes secure connectivity between VPCs and AWS services, preventing exposure of traffic to the internet.
- ❖ PrivateLink allows for the private connection of a VPC with supported AWS services hosted by different AWS accounts.
- ❖ To access AWS services from a VPC using the Gateway VPC endpoint and Interface VPC endpoint.
- ❖ **Gateway endpoints** do not support PrivateLink but allow for connection to Amazon S3 and DynamoDB without the need for an IGW or NAT device in your VPC.
- ❖ For other AWS services, **an interface VPC endpoint** can be created to establish a connection to services through AWS PrivateLink.

AWS PrivateLink

- ❖ Enabling PrivateLink in AWS requires the **creation of an endpoint network interface within the desired subnet**, and assignment of a private IP address from the subnet address range for each specified subnet in the VPC.
- ❖ PrivateLink provides **access to the resources hosted in other VPC or other AWS accounts within the same subnet as the requester** - eliminates the need of NAT gateway, IGW, public IP address, or VPN.
- ❖ AWS PrivateLink enables private connectivity between the **Service Provider** and **Service Consumer** using AWS infrastructure to exchange data without going over the public internet - To achieve this, the **Service Provider** creates an **Endpoint Service** in a private subnet.

AWS PrivateLink

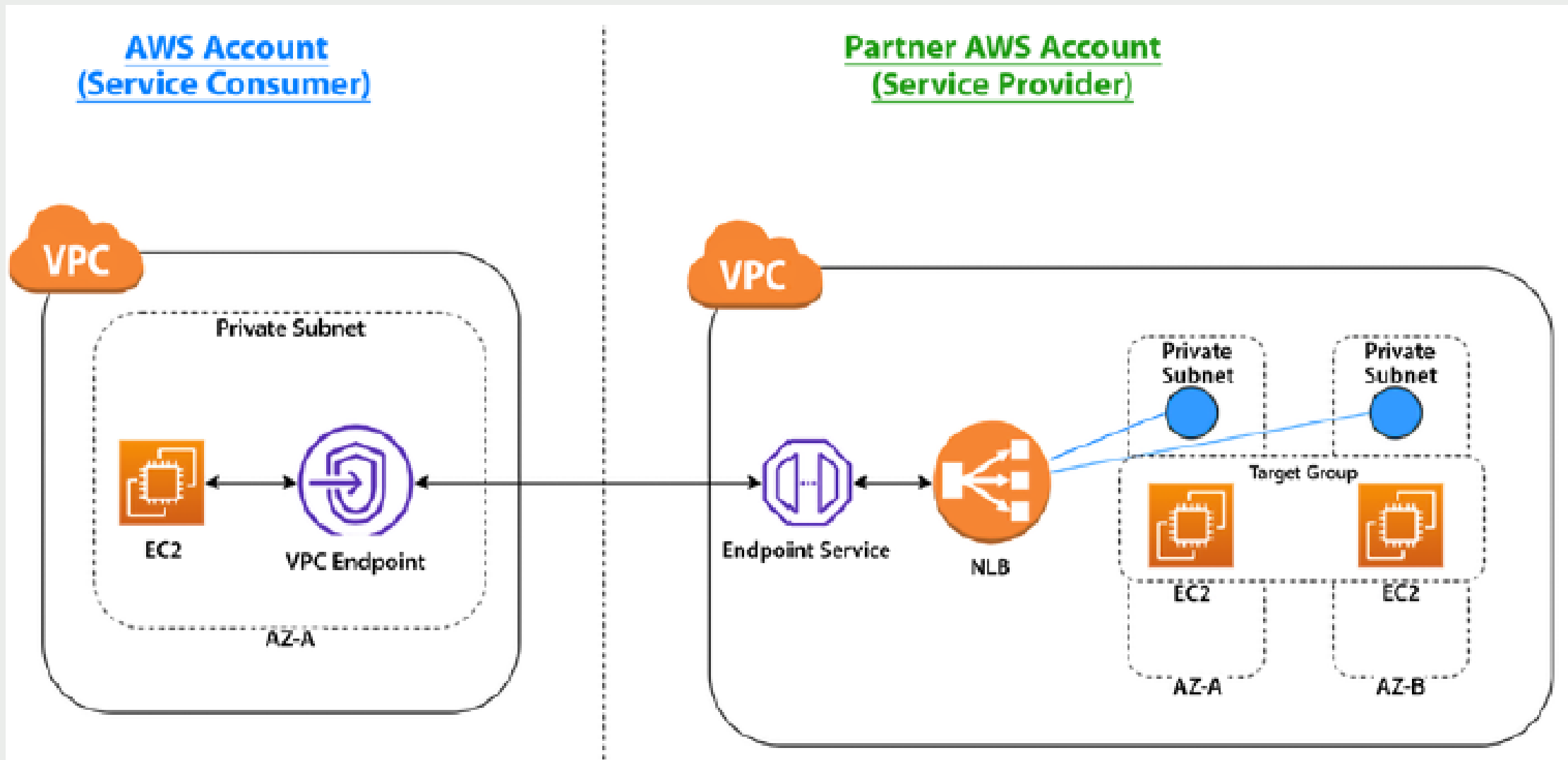


Figure 4.6: PrivateLink between partner Service Provider and Service Consumer accounts

AWS PrivateLink

- ❖ Service Consumer creates an endpoint in a private subnet with the Service **Provider**'s service API as the target.
- ❖ The partner sets up an **Endpoint Service** to expose the service running behind the **load balancer** (**NLB**).
- ❖ An NLB is created in each **Private Subnet**.
- ❖ These services are running on EC2 instances hosted inside a Private Subnet.
- ❖ The client can then create a **VPC Endpoint** with the target as the **Endpoint Service** and use it to consume the service.

AWS PrivateLink

- ❖ Use of PrivateLink between a Service Consumer on an AWS account and an on-premises Service Provider

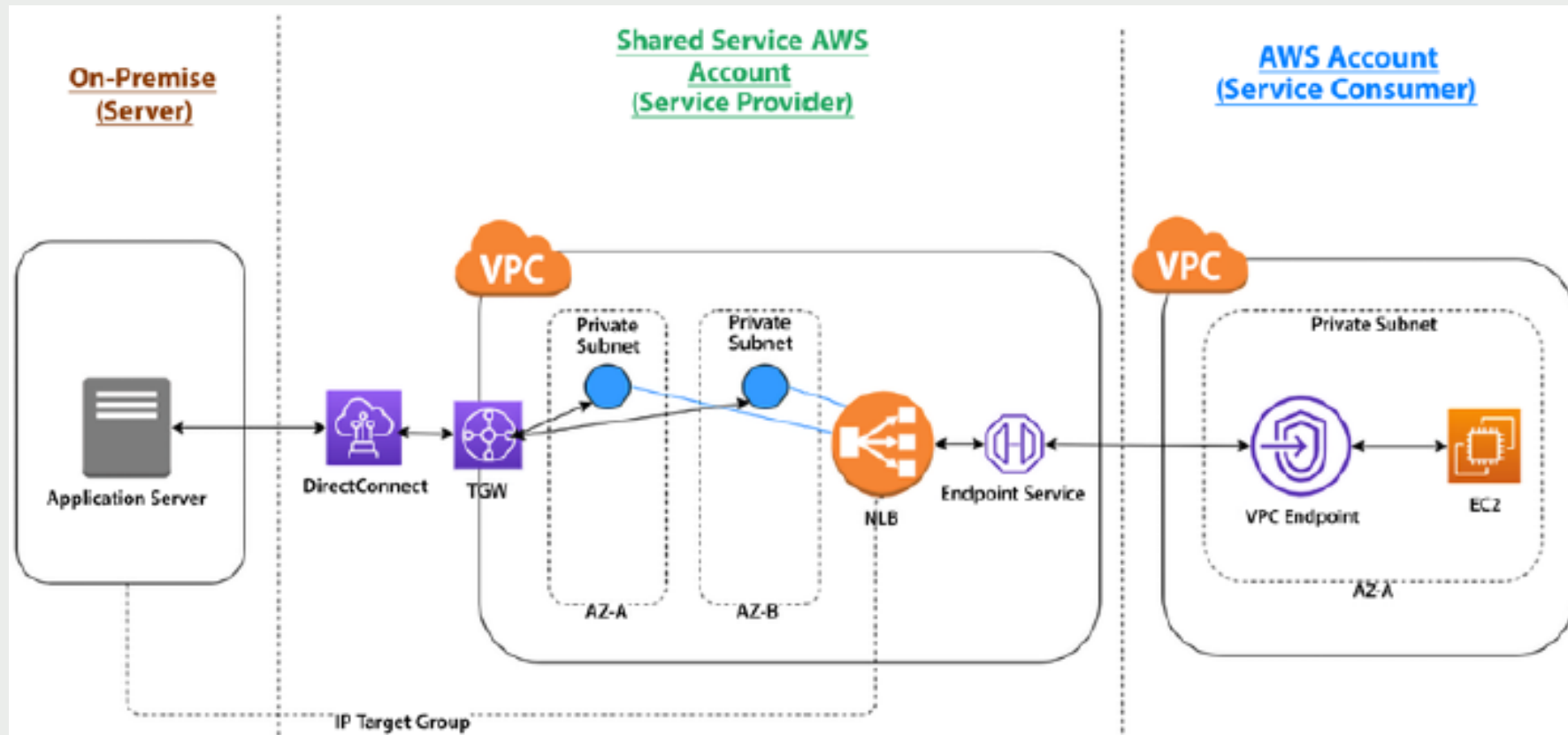


Figure 4.7: PrivateLink between a shared Service Provider, on-premise server, and a Service Consumer account

AWS PrivateLink

- ❖ NLB in the Shared Service account is configured with an auto-scaling group with targets referencing the IP addresses of the on-premise servers.
- ❖ The NLB is then exposed as an **Endpoint Service**.
- ❖ The Service Consumer account can consume this Endpoint Service by creating a **VPC Endpoint**.
- ❖ **DirectConnect** - a dedicated high-speed fiber optics line between the on-premises server and AWS Regions.



Edge networking

Edge Networking

- ❖ Edge networking is like **last-mile delivery** in the supply chain world.
- ❖ Users across the world, from the USA to Australia and India to Brazil, you want each user to have the same experience regardless of the physical location of your server where the application is hosted.
- ❖ There are several components that play their role in building last mile networking.
 - ❖ Route 53
 - ❖ Amazon CloudFront
 - ❖ AWS Global Accelerator (AGA)
 - ❖ AWS Wavelength

Route 53

- ❖ Amazon Route 53 is a **fully managed, simple, fast, secure, highly available, and scalable DNS service**.
- ❖ It provides a reliable and **cost-effective** means for systems and users **to translate names like `www.example.com` into IP addresses like `1.2.3.4`**.
- ❖ Route 53 is a **domain register** where you **can register a new domain**.
- ❖ You can **choose an available domain** and **add it to the cart from the AWS Console** and **define contacts for the domain**.
- ❖ AWS allows you to **transfer your domains to AWS and between accounts**.

Route 53

- ❖ In Route 53, AWS assigns four name servers for all domains, as shown in the screenshot: one for **.com**, one for **.net**, one for **.co.uk**, and one for **.org**.

Route 53 > Hosted zones > examples.com

examples.com [Info](#)

Delete

Test record

Configure query logging

▼ Hosted zone details

Edit

Hosted zone ID

Z0191805YDF6TOZSYX7L

Description

-

Type

Public hosted zone

Record count

2

Name servers

ns-382.awsdns-47.com

ns-987.awsdns-59.net

ns-1555.awsdns-02.co.uk

ns-1249.awsdns-28.org

Route 53

- ❖ Route 53 **supports both public and private hosted zones.**
- ❖ **Public hosted zones** have a route to internet-facing resources and resolve from the internet using global routing policies.
- ❖ Meanwhile, **private hosted zones** have a route to VPC resources and resolve from inside the VPC - helps to integrate with on-premises private zones using forwarding rules and endpoints.
- ❖ Route 53 provides IPv6 support with end-to-end DNS resolution and support for IPv6 forward (AAAA) and reverse (PTR) DNS records, along with health check monitoring for IPv6 endpoints.

Route 53

- ❖ Route 53 provides the following seven types of routing policies for traffic:
 - 1. Simple routing policy** – This is used for a single resource (for example, a web server created for the `www.example.com` website).
 - 2. Failover routing policy** – This is used to configure active-passive failover.
 - 3. Geolocation routing policy** – This routes traffic based on the user's location.
 - 4. Geoproximity routing policy** – This is used for geolocation when users are shifting from one location to another.

Route 53

- ❖ Route 53 provides the following seven types of routing policies for traffic:
 - 5. Latency routing policy** – This optimizes the best latency for the resources deployed in multiple AWS Regions.
 - 6. Multivalue answer routing policy** – This is used to respond to DNS queries with up to eight healthy, randomly selected records.
 - 7. Weighted routing policy** – This is used to route traffic to multiple resource properties as defined by you (for example, you want to say 80% traffic to site A and 20% to site B).

Amazon CloudFront

- ❖ **Amazon CloudFront** is a content delivery service that **accelerates the distribution of both static and dynamic content** like image files, video files, and JavaScript, CSS, or HTML files, **through a network of data centers spread across the globe.** (**data centers are referred to as edge locations**).
- ❖ It used to distribute your content, users requesting content get served by the nearest edge location, providing lower latency and better performance.
- ❖ AWS has over 300 high-density edge locations spread across over 90 cities in 47 countries.
- ❖ All edge locations are equipped with ample cache storage space and intelligent routing mechanisms to increase the edge cache hit ratio.

Amazon CloudFront

- ❖ AWS content distribution edge locations are **connected with high-performance 100 GbE network devices** and are fully redundant, with parallel global networks with default physical layer encryption.
- ❖ Consider an image distribution website, `www.example.com`, hosted in the US, which serves art images.
- ❖ Users can access the URL `www.example.com/art.png`, and the image is loaded.
- ❖ If your server is close to the user, then the image load time will be faster, but if users from other locations like Australia or South Africa, access the same URL, the request has to cross multiple networks before delivering the content to the user's browser.

Amazon CloudFront

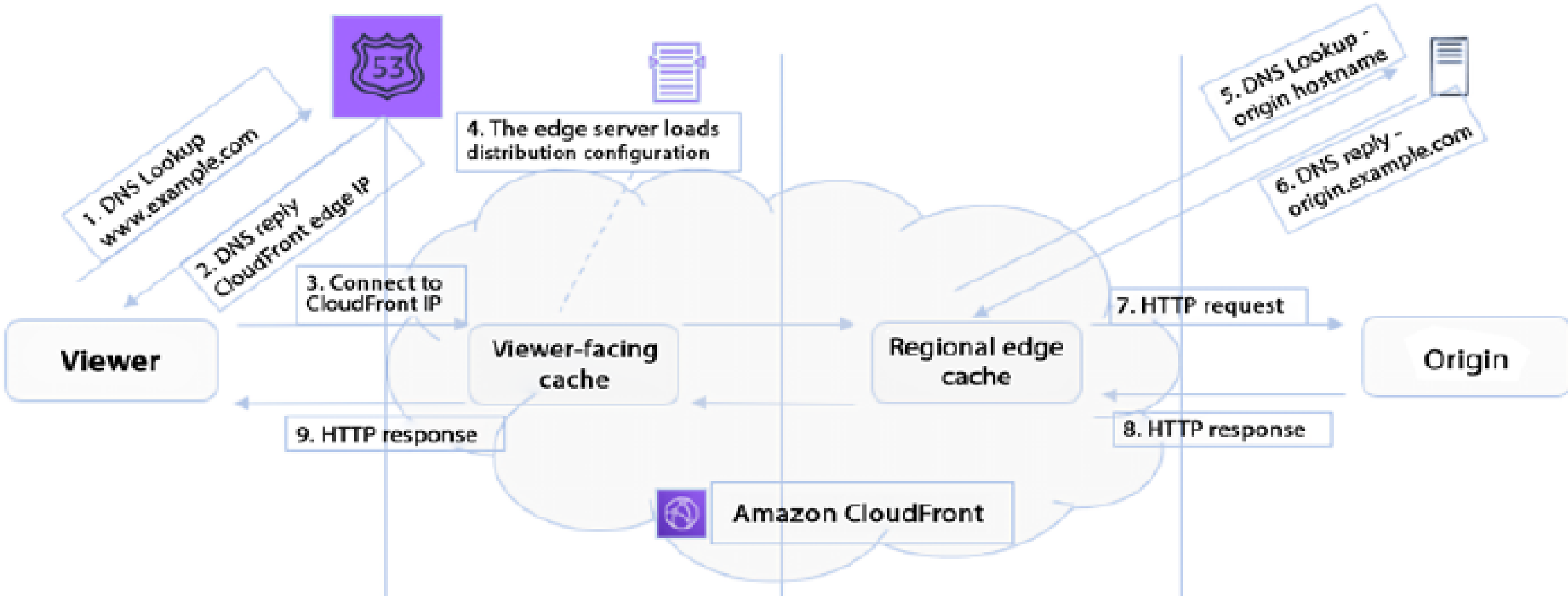


Figure 4.10: HTTP request flow with Amazon CloudFront

Amazon CloudFront

- ❖ As shown in the preceding diagram, when a viewer requests access to page content from the origin server – in this case, `www.example.com` – Route 53 replies with the CloudFront edge IP and redirects the user to the CloudFront location.
- ❖ CloudFront **uses the following rules for content distribution:**
 - ❖ If the **requested content is already in the edge data center**, which means it is a “**cache hit**,” it will be served immediately.
 - ❖ If the **content is not at the edge location** (a “**cache miss**”), CloudFront will request the content from the original location (the web server or S3). The request flows through the AWS backbone, is delivered to the customer, and a copy is kept for future requests.

Amazon CloudFront

- ❖ If you are using CloudFront, it also provides an extra layer of security since your origin server is not directly exposed to the public network.
- ❖ CloudFront eliminates the need to go to the origin server for user requests, and content is served from the nearest location.
- ❖ CloudFront provides security by safeguarding the connection between end-users and the content edge, as well as between the edge network and the origin.
- ❖ By offloading SSL termination to CloudFront, the performance of applications is improved since the burden of processing the required negotiation and SSL handshakes is removed from the origins.

Amazon Global Accelerator (AGA)

- ❖ **AGA** enhances *application availability and performance* by offering fixed static IP addresses as a single entry points, or multiple entry points, to AWS Regions, including ALBs, NLBs, and EC2 instances.
- ❖ **AGA** utilizes the AWS global network to optimize the path from users to applications, thereby improving the performance of TCP and UDP traffic.
- ❖ **AGA** continuously monitors the health of application endpoints and promptly redirects traffic to healthy endpoints within 1 minute, in the event of an unhealthy endpoint detection.

Amazon Global Accelerator (AGA)

- ❖ AGA and CloudFront are distinct services offered by AWS that employ the AWS global network and its edge locations.
- ❖ While CloudFront accelerates the performance of both cacheable (e.g., videos and images) and dynamic (e.g., dynamic site delivery and API acceleration) content, **AGA enhances the performance of various applications over TCP or UDP.**
- ❖ Both services are compatible with AWS Shield, providing protection against DDoS attacks.
- ❖ AGA automatically reroutes your traffic to the nearest healthy endpoint to avoid failure.

Amazon Global Accelerator (AGA)

- ❖ AGA health checks will react to customer backend failure within 30 seconds, which is in line with other AWS load-balancing solutions (such as NLB) and Route 53.
- ❖ AGA raises the bar is with its ability to shift traffic to healthy backends in as short a timeframe as 30 seconds, whereas DNS-based solutions can take minutes to hours to shift the traffic load.
- ❖ Some key reasons to use AGA are:
 - ❖ **Accelerate your global applications** – AGA intelligently directs TCP or UDP traffic from users to the AWS-based application endpoint, providing consistent performance regardless of their geographic location.

Amazon Global Accelerator (AGA)

- ❖ **Improve global application availability** – AGA constantly monitors your application endpoints, including but not limited to ALBs, NLBs, and EC2 instances. It instantly reacts to changes in their health or configuration, redirecting traffic to the next closest available endpoint when problems arise.
- ❖ **Fixed entry point** – AGA provides a set of static IP addresses for use as a fixed entry point to your AWS application. Announced via anycast and delivered from AWS edge locations worldwide, these eliminate the complexity of managing the IP addresses of multiple endpoints and allow you to scale your application and maintain DDoS resiliency with AWS Shield.

Amazon Global Accelerator (AGA)

- ❖ **Protect your applications** – AGA allows you to serve internet users while keeping your ALBs and EC2 instances private.
- ❖ **AGA allows customers to run global applications in multiple AWS Regions.**
- ❖ Traffic destined to static IPs is globally distributed, and end user requests are ingested through AWS's closest edge location and routed to the correct regional resource for better availability and latency.
- ❖ This **global endpoint supports TCP and UDP and does not change** even as customers move resources between Regions for failover or other reasons.

AWS Wavelength

- ❖ **AWS Wavelength** is designed to reduce network latency when connecting to applications from 5G-connected devices by providing infrastructure deployments within the Telco 5G network service providers' data centers.
- ❖ It **allows application traffic to reach application servers running in Wavelength Zones**, as well as AWS compute and storage services.
- ❖ This eliminates the need for traffic to go through the internet, which can introduce latency of up to 10s of milliseconds and limit the full potential of the bandwidth and latency advancements of 5G.

AWS Wavelength

- ❖ AWS Wavelength allows for the creation and implementation of real-time, low-latency applications, such as edge inference, smart factories, IoT devices, and live streaming.
- ❖ **Benefits of AWS Wavelength** are:
 - ❖ **Ultra-low latency for 5G** – Wavelength combines the AWS core services, such as compute and storage, with low-latency 5G networks. It helps you to build applications with ultra-low latencies using the 5G network.
 - ❖ **Consistent AWS experience** – You can use the same AWS services you use daily on the AWS platform.

AWS Wavelength

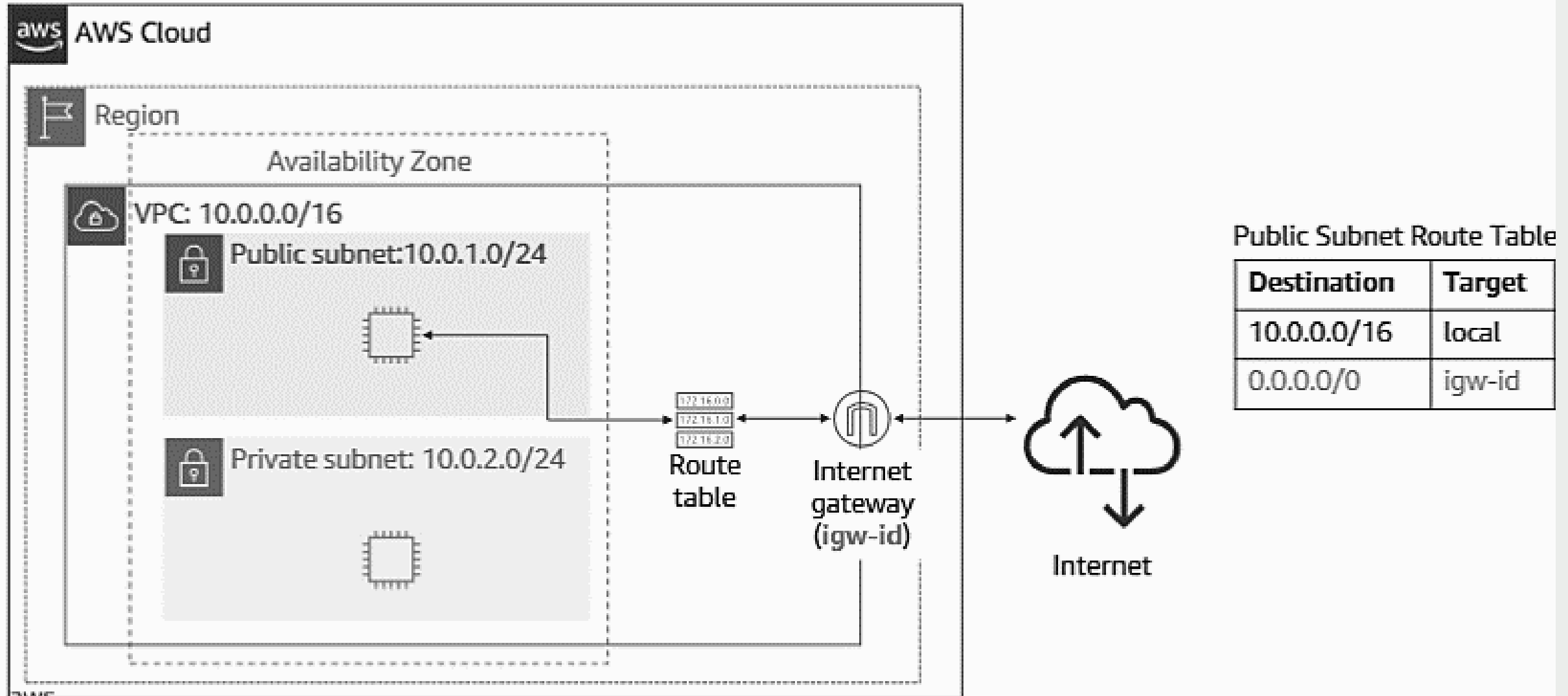
- ❖ **Benefits of AWS Wavelength are:**

- ❖ **Global 5G network** – Wavelength is available in popular Telco networks such as Verizon, Vodafone, and SK Telecom across the globe, including the US, Europe, Korea, and Japan, which enables ultra-low latency applications for a global user base.
- ❖ **Wavelength Zones are connected to a Region and provide access to AWS services.**
- ❖ **Architecting edge applications using a hub-and-spoke model with the Region is recommended for scalable and cost-effective options for less latency-sensitive applications.**



Example: Routing traffic using VPC

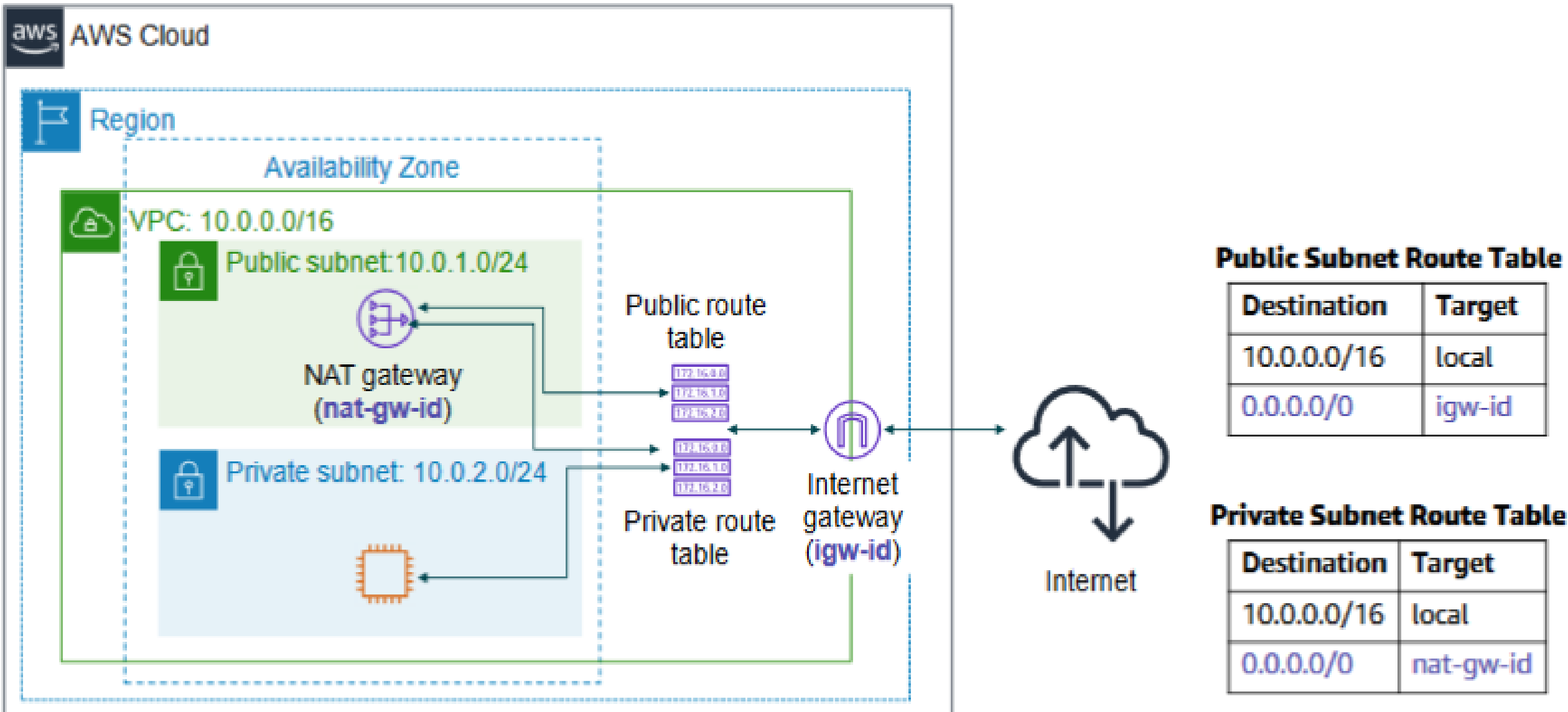
Internet gateway



Internet gateway

- ❖ An internet gateway is a scalable, redundant, and highly available VPC component that allows communication between instances in your VPC and the internet.
- ❖ An internet gateway serves two purposes: to provide a target in your VPC route tables for internet-routable traffic, and to perform network address translation for instances that were assigned public IPv4 addresses.
- ❖ To make a subnet public, you attach an internet gateway to your VPC and add a route to the route table to send non-local traffic through the internet gateway to the internet (0.0.0.0/0).

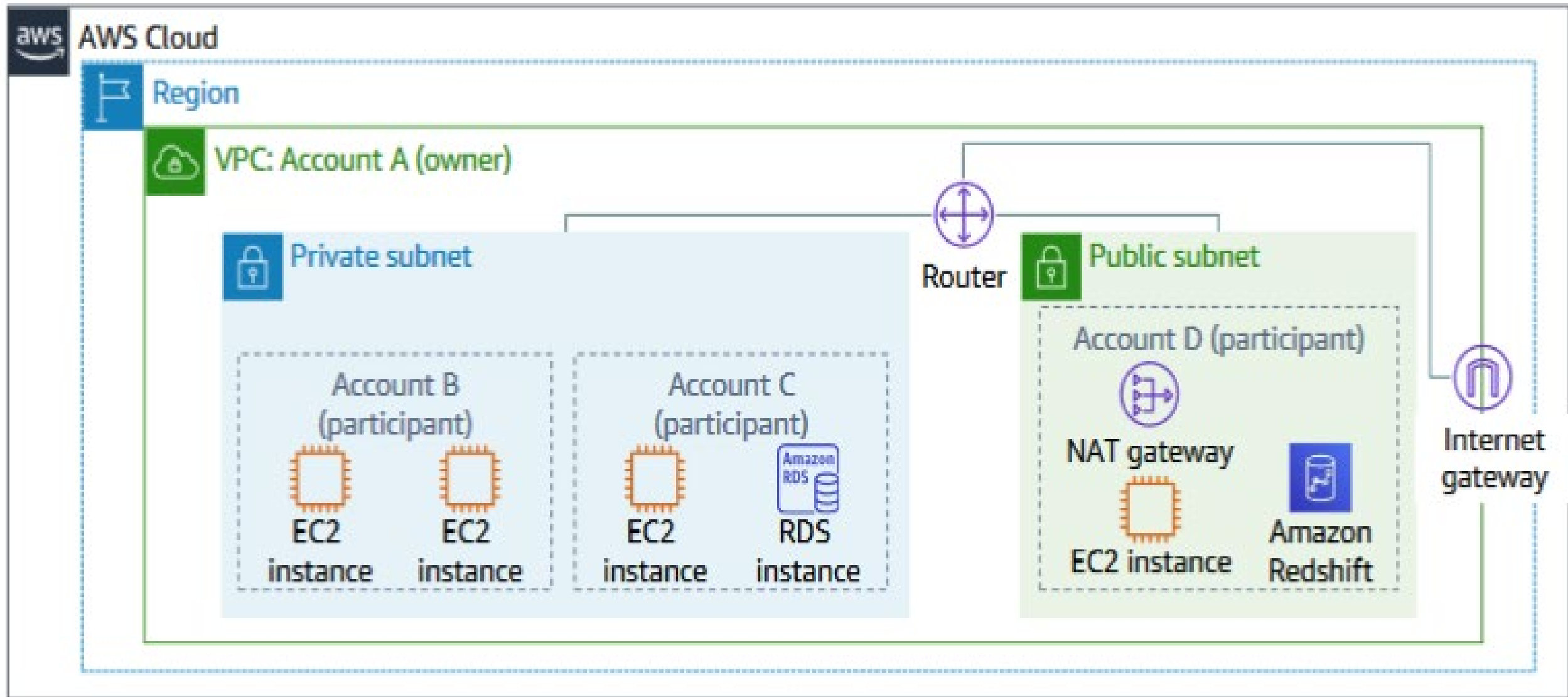
Network address translation (NAT) gateway



Network address translation (NAT) gateway

- ❖ It enables instances in a **private subnet to connect to the internet or other AWS services**, but prevents the internet from initiating a connection with those instances.
- ❖ To **create a NAT gateway**, specify the **public subnet in which the NAT gateway should reside**, and must also specify an Elastic IP address to associate with the NAT gateway when you create it.
- ❖ After creation of NAT gateway, **update the route table** that is **associated with one or more of your private subnets** to point internet-bound traffic to the NAT gateway. (So, your private subnets can communicate with the internet)
- ❖ NAT instance in a public subnet - **better availability, higher bandwidth, and less administrative effort.**

VPC sharing



VPC sharing

- ❖ It **enables customers to share subnets with other AWS accounts** in the same organization in AWS Organizations.
- ❖ It **enables multiple AWS accounts to create their application resources**—such as Amazon EC2 instances, Amazon Relational Database Service (Amazon RDS) databases, Amazon Redshift clusters, and AWS Lambda functions—into shared, centrally managed VPCs.
- ❖ In this model, the account that owns the VPC (owner) shares one or more subnets with other accounts (participants) that belong to the same organization in AWS Organizations.
- ❖ After a subnet is shared, the participants can view, create, modify, and delete their application resources in the subnets that are shared with them.

VPC sharing

- ❖ Participants cannot view, modify, or delete resources that belong to other participants or the VPC owner.
- ❖ **VPC sharing offers several benefits:**
 - ❖ **Separation of duties** –Centrally controlled VPC structure, routing, IP address allocation
 - ❖ **Ownership** –Application owners continue to own resources, accounts, and security groups
 - ❖ **Security groups** –VPC sharing participants can reference the security group IDs of each other
 - ❖ **Efficiencies** –Higher density in subnets, efficient use of VPNs and AWS Direct Connect

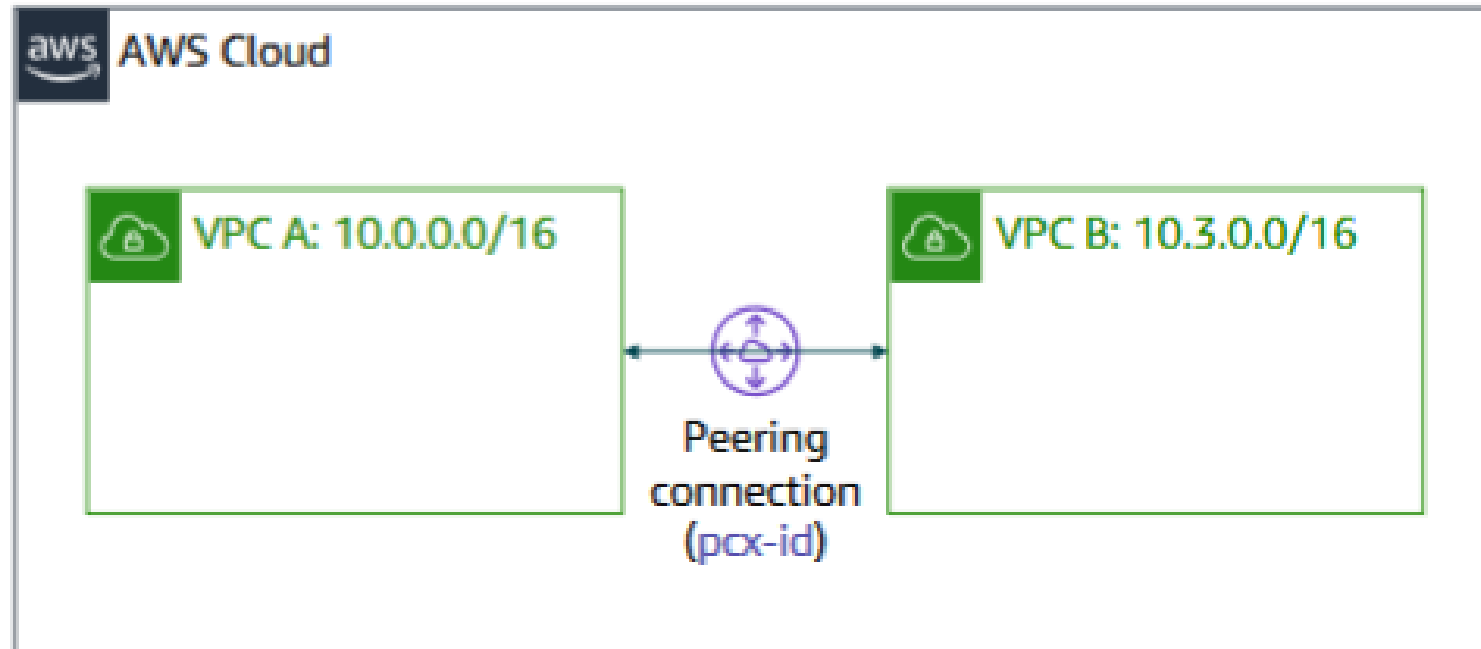
VPC sharing

- ❖ VPC sharing offers several benefits:
 - ❖ **No hard limits** –Hard limits can be avoided—for example, 50 virtual interfaces per AWS Direct Connect connection through simplified network architecture
 - ❖ **Optimized costs** –Costs can be **optimized through the reuse of NAT gateways, VPC interface endpoints, and intra-Availability Zone traffic VPC sharing** - enables you to decouple accounts and networks.

VPC peering

- ❖ A **VPC peering** connection is a networking connection between two VPCs that enables you to route traffic between them privately.
- ❖ Instances in either VPC can communicate with each other as if they are within the same network.
- ❖ Creating a VPC peering connection between your own VPCs, with a VPC in another AWS account, or with a VPC in a different AWS Region.
- ❖ Setting up the peering connection, **must create rules in your route table to allow the VPCs to communicate with each other through the peering resource.**

VPC peering



Route Table for VPC A

Destination	Target
10.0.0.0/16	local
10.3.0.0/16	pcx-id

Route Table for VPC B

Destination	Target
10.3.0.0/16	local
10.0.0.0/16	pcx-id

You can connect VPCs in your own AWS account, between AWS accounts, or between AWS Regions.

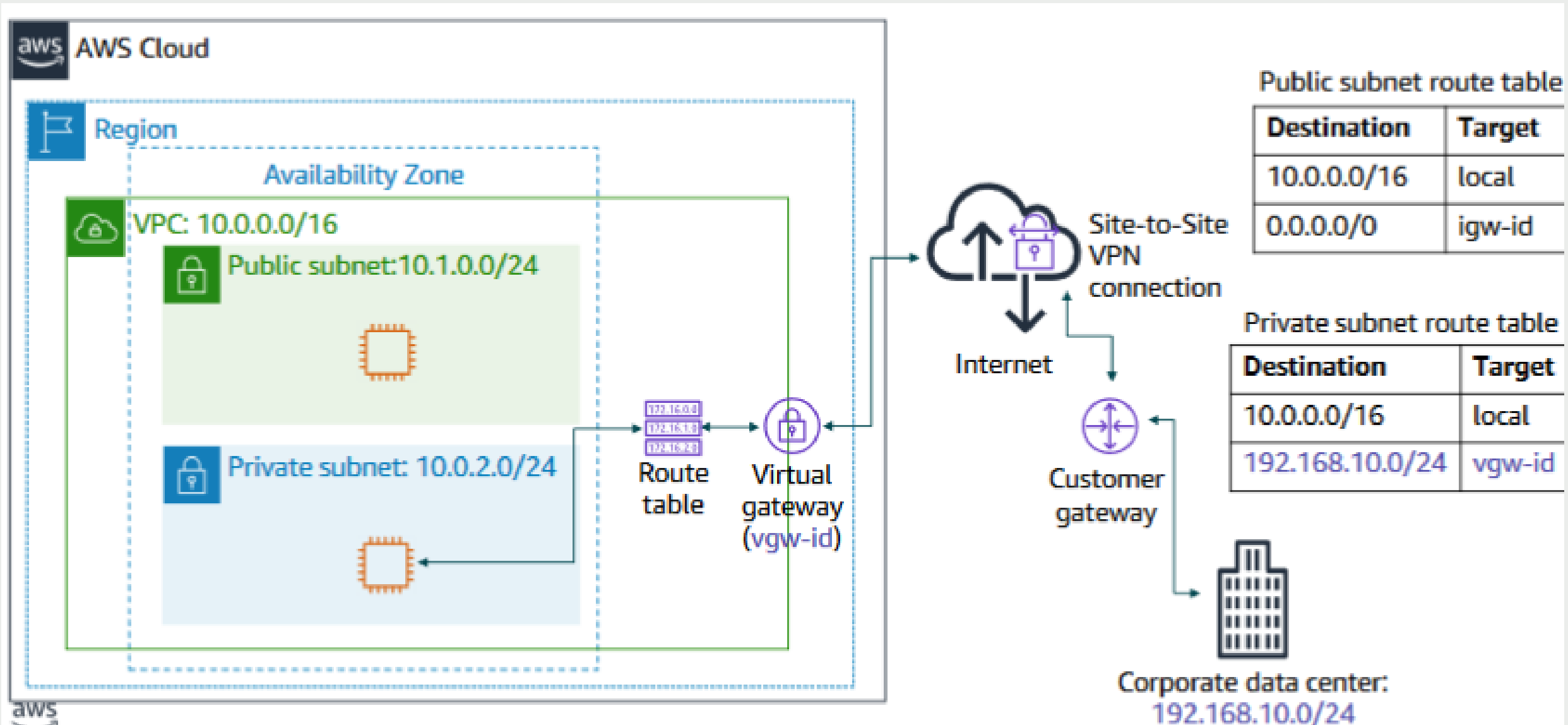
Restrictions:

- IP spaces cannot overlap.
- Transitive peering is not supported.
- You can only have one peering resource between the same two VPCs.

VPC peering

- ❖ For example, suppose that you have two VPCs. In the route table for VPC A, you set the destination to be the IP address of VPC B and the target to be the peering resource ID. In the route table for VPC B, you set the destination to be the IP address of VPC A and the target to be the peering resource ID.
- ❖ **VPC peering has some restrictions:**
 - ❖ *IP address ranges **cannot overlap.***
 - ❖ ***Transitive peering is not supported.** For example, suppose that you have three VPCs: A, B, and C. VPC A is connected to VPC B, and VPC A is connected to VPC C. However, VPC B is not connected to VPC C implicitly. To connect VPC B to VPC C, you must explicitly establish that connectivity.*
 - ❖ ***Only one peering resource between the same two VPCs.***

AWS Site-to-Site VPN



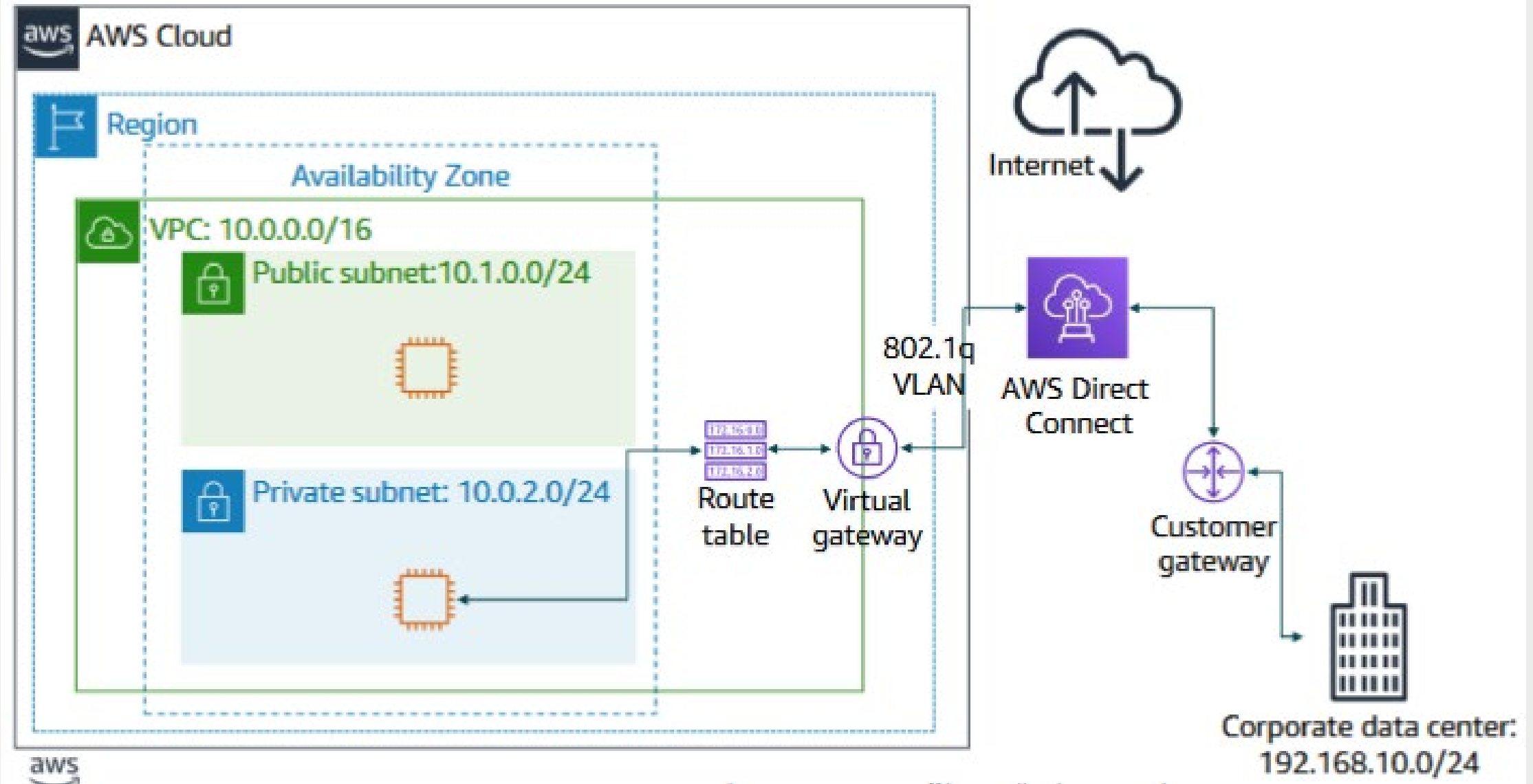
AWS Site-to-Site VPN

- ❖ **By default, instances that you launch into a VPC cannot communicate with a remote network.**
- ❖ To connect your VPC to your remote network (that is, create a virtual private network or VPN connection), you:
 1. **Create a new virtual gateway device** (called a virtual private network (VPN) gateway) and **attach it to your VPC.**
 2. **Define the configuration of the VPN device or the customer gateway.** The customer gateway is not a device but an AWS resource that provides information to AWS about your VPN device.

AWS Site-to-Site VPN

- 3. Create a custom route table to point corporate data center-bound traffic to the VPN gateway.** You also must update security group rules. (You will learn about security groups in the next section.)
- 4. Establish an AWS Site-to-Site VPN (Site-to-Site VPN) connection to link the two systems together.**
- 5. Configure routing to pass traffic through the connection.**

AWS Direct Connect



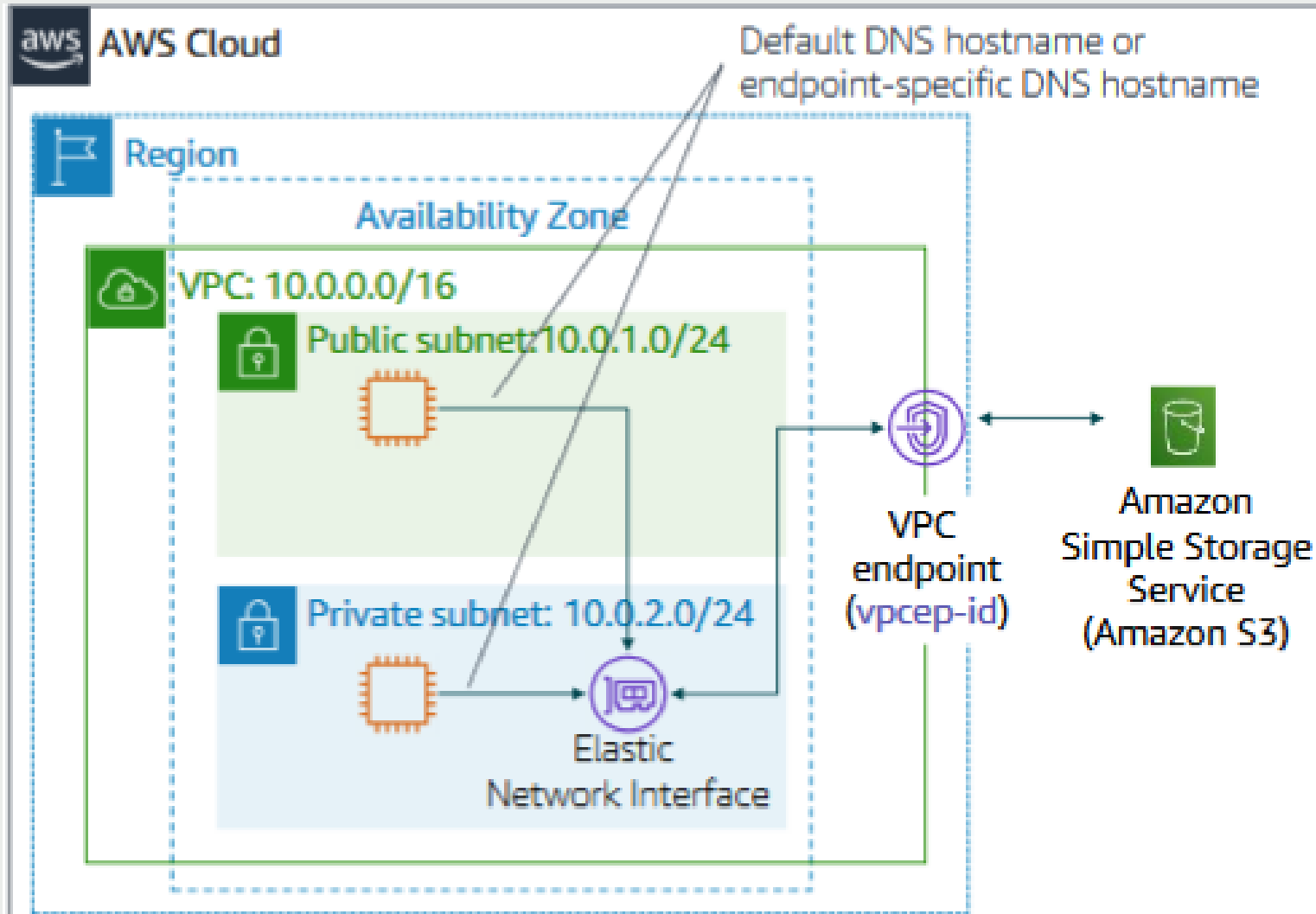
AWS Direct Connect

- ❖ One of the challenges of network communication is network performance.
- ❖ Performance can be negatively affected if your data center is located far away from your AWS Region.
- ❖ For such situations, **AWS offers AWS Direct Connect, or DX.**
- ❖ It enables you to establish a dedicated, private network connection between your network and one of the DX locations.
- ❖ This private connection can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than internet-based connections.
- ❖ **DX uses open standard 802.1q VLANs.**

VPC endpoints

- ❖ A VPC endpoint **is a virtual device** that enables you to **privately connect your VPC to** supported **AWS services** and VPC endpoint services that **are powered by AWS PrivateLink**.
- ❖ Connection to these services does not require an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection.
- ❖ Instances in your VPC do not require public IP addresses to communicate with resources in the service.
- ❖ Traffic between your VPC and the other service does not leave the Amazon network.

VPC endpoints



Public Subnet Route Table

Destination	Target
10.0.0.0/16	local
Amazon S3 ID	vpcep-id

Two types of endpoints:

- **Interface** endpoints (powered by AWS PrivateLink)
- **Gateway** endpoints (Amazon S3 and Amazon DynamoDB)

VPC endpoints

- ❖ There are two types of VPC endpoints:
- ❖ An **interface VPC endpoint** (interface endpoint) enables you to connect to services that are powered by AWS PrivateLink.
 - ❖ These services include some AWS services, services that are hosted by other AWS customers and AWS Partner Network (APN) Partners in their own VPCs (referred to as endpoint services), and supported AWS Marketplace APN Partner services.
 - ❖ The owner of the service is the service provider, and you—as the principal who creates the interface endpoint—are the service consumer.
 - ❖ You are charged for creating and using an interface endpoint to a service. Hourly usage rates and data processing rates apply.
- ❖ **Gateway endpoints:** The use of gateway endpoints incurs no additional charge.
 - ❖ *Standard charges for data transfer and resource usage apply.*

AWS Transit Gateway

- ❖ You can configure your VPCs in several ways, and take advantage of numerous connectivity options and gateways. These options and gateways include **AWS Direct Connect (via DX gateways)**, **NAT** gateways, internet gateways, VPC peering, etc.
- ❖ It is not uncommon to find AWS customers with hundreds of VPCs distributed across AWS accounts and Regions to serve multiple lines of business, teams, projects, and so forth.
- ❖ Things get more complex when customers start to set up connectivity between their VPCs. All the **connectivity options are strictly point-to-point**, so the number of VPC-to-VPC connections can grow quickly.

AWS Transit Gateway

- ❖ As you grow the number of workloads that run on AWS, you must be able to scale your networks across multiple accounts and VPCs to keep up with the growth.
- ❖ Though you can use VPC peering to connect pairs of VPCs, managing point-to-point connectivity across many VPCs without the ability to centrally manage the connectivity policies can be operationally costly and difficult.
- ❖ For on-premises connectivity, you must attach your VPN to each individual VPC. This solution can be time-consuming to build and difficult to manage when the number of VPCs grows into the hundreds.

AWS Transit Gateway

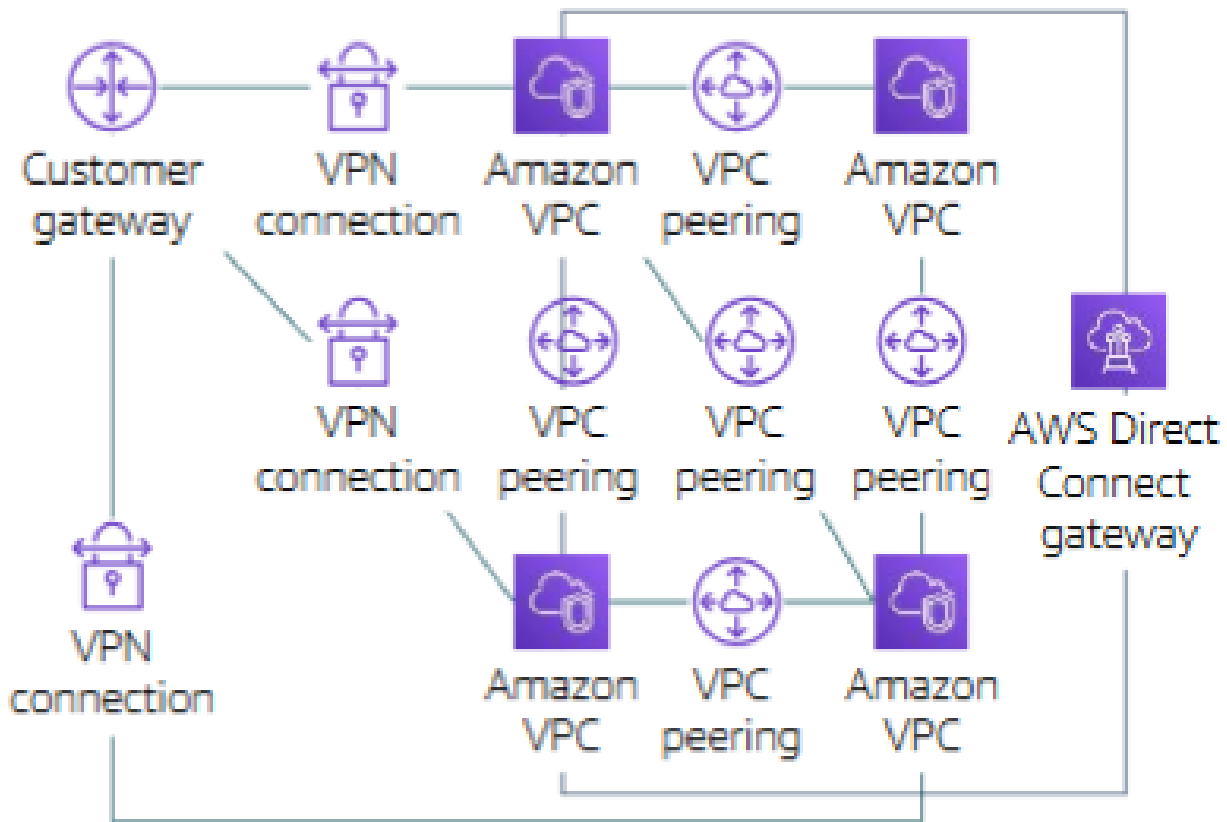
- ❖ To solve this problem, you can use AWS Transit Gateway to simplify your networking model.
- ❖ With AWS Transit Gateway, you only need to create and manage a single connection from the central gateway into each VPC, on-premises data center, or remote office across your network.
- ❖ A transit gateway acts as a hub that controls how traffic is routed among all the connected networks, which act like spokes.
- ❖ This hub-and-spoke model significantly simplifies management and reduces operational costs because each network only needs to connect to the transit gateway and not to every other network.

AWS Transit Gateway

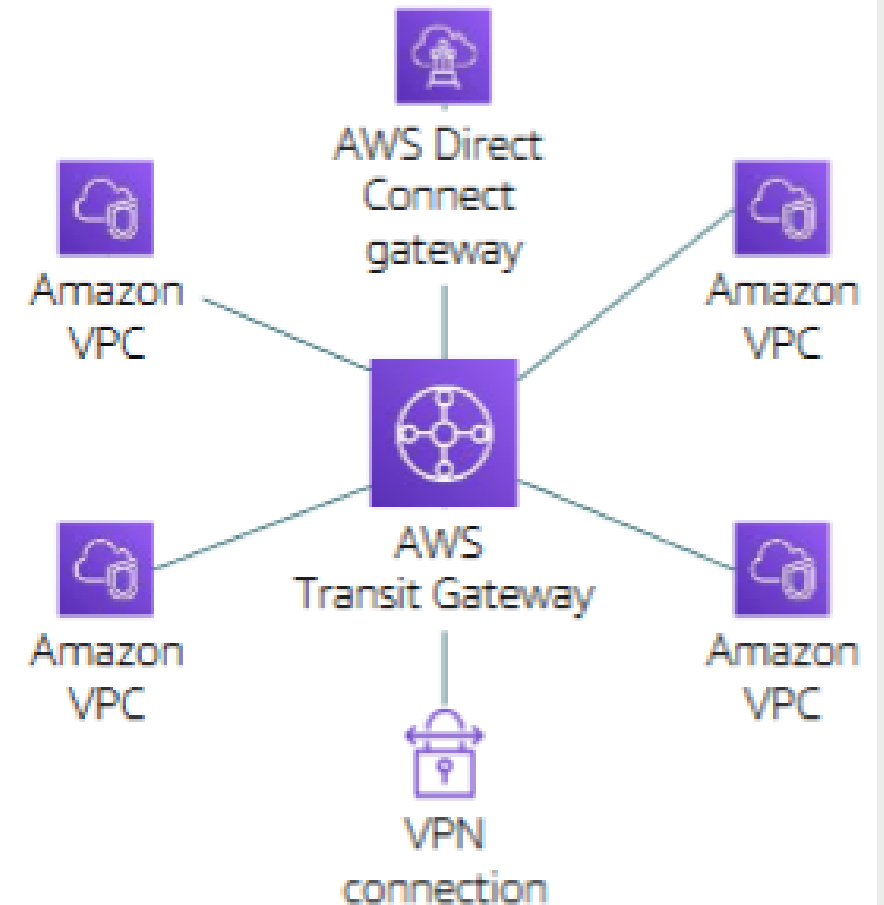
- ❖ Any new VPC is connected to the transit gateway, and is then automatically available to every other network that is connected to the transit gateway.
- ❖ This ease of connectivity makes it easier to scale your network as you grow.

AWS Transit Gateway

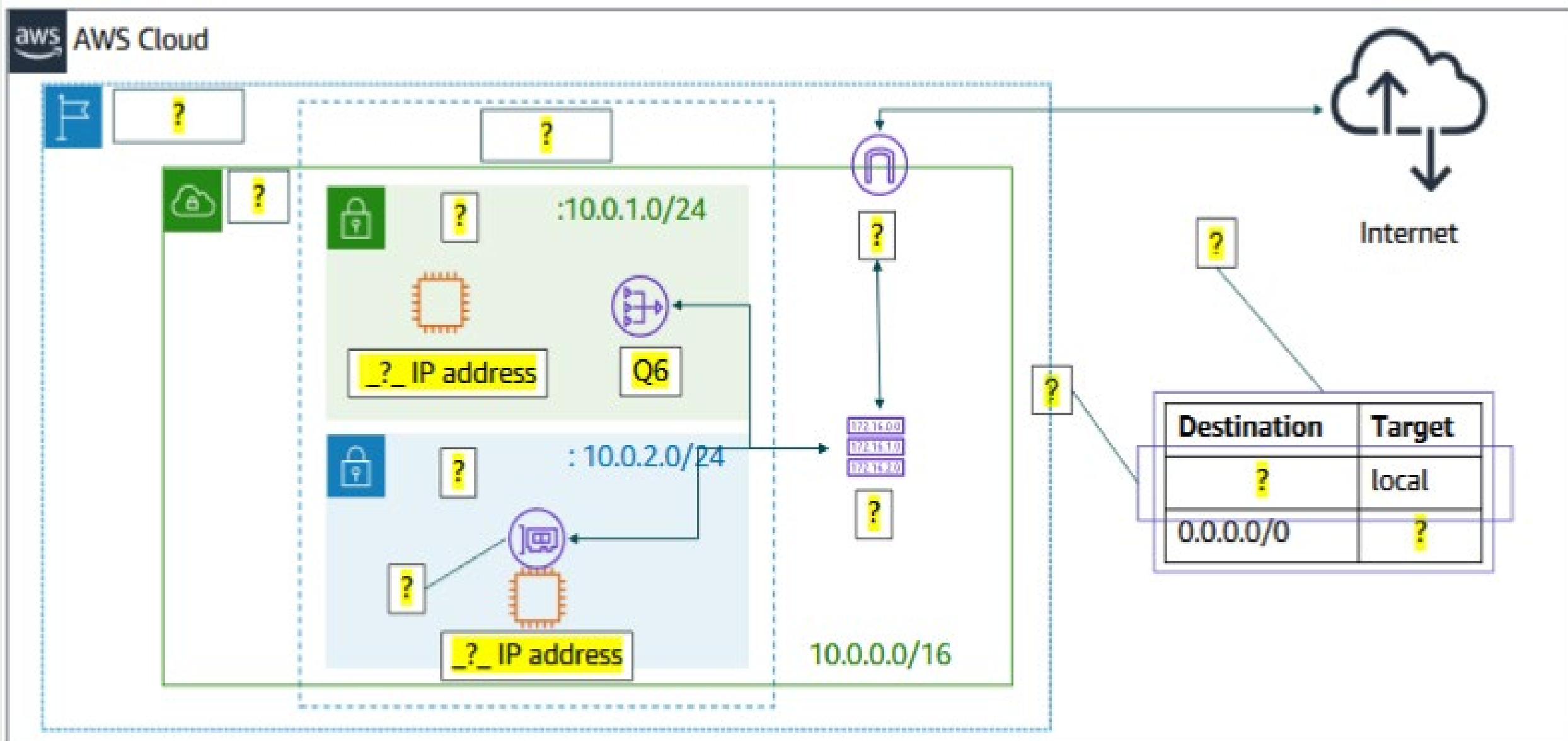
From this...



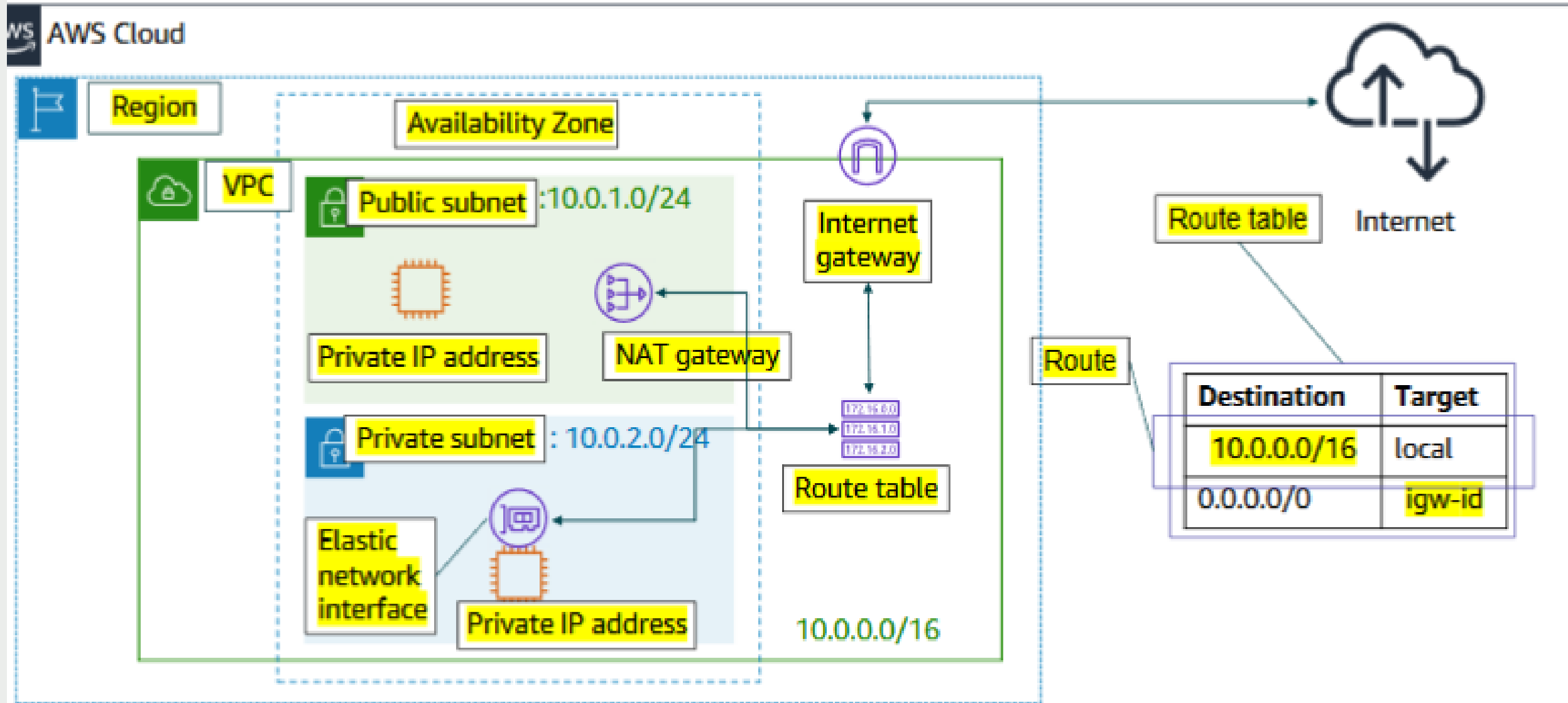
To this...



Label this network diagram



Label this network diagram



THANK YOU

Module 3

BCSE355L – Aws Solution Architect